

Effective Force Laws from Phase Geometry

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Abstract

We present a comprehensive derivation of all known effective force laws as emergent geometric phenomena within Phase Theory (Hanks 2026), a foundational framework in which the sole fundamental constituent of physical reality is a phase-bearing substrate $\Phi : M \rightarrow T$, where M is a pre-geometric topological base space and $T = U(1) \times SU(2) \times SU(3)$ is the target manifold. Forces, in this framework, are not fundamental interactions mediated by exchange particles; they are emergent approximations to defect motion along admissibility gradients in phase geometry. The central result of this paper is a complete derivation of the effective force law $F_{\text{eff}} = m\nabla A$ from the variational structure of the phase-inconsistency functional $I[\Phi] = \int_M \rho(\Phi, D\Phi) d\mu$, where $A[\Phi] = A_{\text{max}} - I[\Phi]$ is the complementary admissibility functional.

We establish the following principal results. (1) The inverse-square law emerges universally from the geometric dilution of admissibility influence in three emergent spatial dimensions: isotropic admissibility flux from a localized defect yields gradient magnitude scaling as r^{-2} (Proposition 42.2). (2) Potential energy is identified as admissibility bookkeeping: $V_{\text{eff}}(\mathbf{x}) \equiv -A(\mathbf{x}) + \text{const}$ whenever the admissibility gradient is conservative. (3) The superposition principle for forces is exact in the linearized phase dynamics regime and acquires nonlinear corrections at third order in the phase strain. (4) Attraction and repulsion between defects are uniquely determined by topological compatibility: complementary topological invariants produce admissibility basins (attraction); identical invariants produce admissibility ridges (repulsion). (5) Force range is governed by phase-mode mass: massless modes yield power-law (long-range) forces; massive modes yield exponentially screened (short-range) forces, recovering Yukawa potentials. (6) Newton's three laws are reinterpreted as low-curvature-limit consequences of admissibility gradient dynamics. (7) Relativistic corrections to force laws follow from the covariant geodesic equation in the emergent metric $g_{\mu\nu}(\mathbf{x}) = \alpha \langle D_\mu \Phi \cdot D_\nu \Phi \rangle_{\text{coh}}$. (8) The Lorentz force law is uniquely determined by U(1) gauge invariance of the admissibility functional applied to moving charged defects. (9) We prove Theorem 42.1 (No Force Carrier Theorem): no degree of freedom of the phase substrate is exchanged between defects during an effective force interaction. Virtual particles in Feynman diagrams are perturbative approximations to admissibility landscape geometry. (10) All four forces unify at the phase unification scale $\Lambda_{\text{PU}} \sim 10^{15}\text{--}10^{16}$ GeV, at which the symmetry-breaking pattern of the phase vacuum is restored. We derive specific experimental predictions distinguishable from Standard Model and General Relativity predictions and enumerate protocols for their detection. This paper is Paper 42 in the Phase Theory series; it supersedes earlier treatments of force emergence in Papers 9, 38, and 41 and provides the theoretical foundation for Papers 43 (Post-Newtonian Phase Corrections) and 44 (Gauge Theory from Phase Topology).

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1. Introduction and Motivation

1.1 The Historical Concept of Force

The concept of force has been central to physics since Newton's formulation of classical mechanics in 1687 [1]. Newton introduced force as a primitive — an entity that causes change in the state of motion of a body — and encoded its behavior in three laws and the inverse-square gravitational attraction. For two centuries, this framework proved extraordinarily successful, describing planetary orbits, tides, projectile motion, and the dynamics of rigid bodies with exquisite precision. Yet Newton himself remained deeply troubled by the ontological status of gravitational force, famously declining to "feign hypotheses" about its mechanism: *Hypotheses non fingo*.

Maxwell's unification of electricity and magnetism in 1865 [2] elevated the field to a fundamental role: forces are mediated by fields, which carry energy and momentum independently of their sources. The electromagnetic field possesses its own dynamics, embodied in Maxwell's equations, and the force on a charged body is a local consequence of the field value at the body's location. This was a decisive conceptual advance — forces became emergent from field configurations — yet the field itself was elevated to a new primitive, and the question of what a field *is* was deferred.

Einstein's general relativity (1915) [3] executed a second conceptual revolution: gravitational force was reinterpreted as the curvature of spacetime geometry. In GR, freely falling bodies experience no force at all; they traverse geodesics in a curved manifold. "Gravity" as experienced on Earth's surface is the reaction force that prevents free fall. This was the first complete dissolution of a force into geometry, and it pointed the way toward the program undertaken in the present paper.

The development of quantum field theory in the twentieth century [4, 5, 6] extended the field-mediation picture to the quantum realm. Forces arise from the exchange of virtual gauge bosons: the photon mediates electromagnetism, the $W^\pm$ and Z^0 bosons mediate the weak nuclear force, and gluons mediate the strong force. The Yang-Mills framework [7] unified this picture mathematically, grounding all gauge forces in local symmetry principles. The Standard Model of particle physics [8, 9, 10] encodes all known non-gravitational forces within the gauge group $U(1) \times SU(2) \times SU(3)$ and has been confirmed to extraordinary precision.

1.2 The Unsatisfying Primitiveness of Force Mediators

Despite the triumph of the Standard Model, the force-mediation picture carries deep unsatisfying features. First, it requires the postulation of 12 distinct gauge bosons (1 photon + 3 weak + 8 gluons) plus the graviton, each

introduced to "explain" a distinct force. The proliferation of mediators merely relocates the explanatory burden: instead of explaining force, we must explain why these specific mediators exist, why they carry the quantum numbers they do, and why the gauge group is $U(1) \times SU(2) \times SU(3)$ rather than some other group. Second, the problem of unification — the aspiration to derive all forces from a single underlying principle — remains unsolved within the Standard Model. Grand Unified Theories (GUTs) reduce the gauge group but do not dissolve forces into geometry. String theory attempts unification through higher-dimensional geometry but has not yielded experimentally confirmed predictions. Third, and most fundamentally, the virtual-particle picture of force mediation is a perturbative approximation. Virtual particles do not exist in any empirically verifiable sense; they are bookkeeping devices in a perturbative expansion that may itself be only asymptotically valid.

These difficulties motivate the central program of Phase Theory [11]: to dissolve force as a primitive by deriving all force phenomenology from a single underlying geometric principle — the minimization of phase inconsistency in a topologically structured substrate. In Phase Theory, forces are not fundamental. They are emergent tendencies of constrained phase transport along admissibility gradients. Nothing pulls; nothing pushes; phase geometry prefers.

1.3 Phase Theory's Reframing

Phase Theory proposes that the sole fundamental constituent of physical reality is a phase-bearing substrate $\Phi : M \rightarrow T$, where M is a pre-geometric topological base space (a directed acyclic graph at the microscopic level) and $T = U(1) \times SU(2) \times SU(3)$ is the target manifold encoding all internal symmetry structure [11, 12]. Spacetime, matter, geometry, and force are all emergent from the global organization and consistency of Φ . The dynamics of Φ are governed by the phase-inconsistency functional $I[\Phi] = \int_M \rho(\Phi, D\Phi) d\mu$, which is minimized subject to topological constraints.

In this picture, a "force" between two particles (topological defects in Φ , introduced in Paper 9 [13]) is the gradient of the admissibility functional $A[\Phi] = A_{\max} - I[\Phi]$ with respect to defect position. When one defect is present, it distorts the phase substrate around it; a second defect moving through this distorted substrate experiences a tendency to move in the direction of increasing admissibility. This tendency is what we call a force. No exchange of any substrate degree of freedom is required; the force is entirely a consequence of the geometry of the admissibility landscape.

1.4 Structure and Scope of This Paper

This paper is Paper 42 in the Phase Theory series. It builds on the mathematical framework established in Paper 5 [12], the particle defect theory of Paper 9 [13], the mass and motion theory of Paper 38 [14], and the reconfiguration interaction theory of Paper 41 [15]. The present paper provides a complete, systematic derivation of all effective force laws as consequences of admissibility gradient geometry. We organize the paper as follows. Sections 2–6 review Phase Theory foundations and establish the necessary technical machinery. Sections 7–16 derive the general force law and its consequences, including the inverse-square law (Section 10), Coulomb's law (Section 11), Newtonian gravity (Section 12), angular momentum conservation (Section 13), attraction/repulsion topology (Section 14), and force range classification (Section 15). Sections 17–24 derive relativistic and quantum corrections, including the Lorentz force (Section 20), strong force (Section 21), weak force (Section 22), and post-Newtonian gravity (Section 23). Sections 25–33 address advanced topics: effective potential catalog (Section 25), force description breakdown (Section 26), the No-Force-Carrier Theorem (Section 27), QFT comparison (Section 28), force unification (Section 29), dark forces (Section 30), Casimir effect (Section 31), tidal forces (Section 32), and curved-geometry force laws (Section 33). Sections 34–39 discuss experimental signatures, protocols, comparisons with

prior work, limitations, and conclusions. Five appendices provide formal proofs, catalogs, and reference material. We use natural units $\hbar = c = 1$ throughout unless otherwise stated.

2. Review of Phase Theory Foundations

2.1 The Fundamental Postulate

Phase Theory rests on a single ontological postulate, introduced in Paper 1 [16]: *The sole fundamental constituent of physical reality is a phase configuration Φ , which is a map from a pre-geometric topological base space M to a target manifold T .* All structures — particles, fields, spacetime, forces — are emergent patterns in Φ . This postulate is more radical than those of string theory or loop quantum gravity in that it does not presuppose spacetime, matter, or any specific dynamical law beyond phase consistency. The target manifold $T = U(1) \times SU(2) \times SU(3)$ is identified with the gauge structure of the Standard Model, but in Phase Theory this identification is not postulated — it is (in principle) derivable from the topological structure of stable defects (a program underway in Papers 30–35).

2.2 The Five Axioms of Phase Theory

Phase Theory is axiomatized in Paper 5 [12] through five foundational axioms, which we now recall.

Axiom 1 (Global Phase Consistency).

The realized phase configuration Φ^* is the global minimizer of the phase-inconsistency functional $I[\Phi]$ over the space Γ of topologically admissible configurations:

$$\Phi^* = \arg \min_{\Phi \in \Gamma} I[\Phi].$$

Axiom 2 (Update Ordering).

The base space M carries the structure of a directed acyclic graph (DAG) encoding the causal order of phase updates. Spacetime dimension and causal structure are emergent from the large-scale statistics of this DAG. This axiom implies that there is no fundamental time; temporal ordering is the emergent manifestation of DAG structure.

Axiom 3 (Topological Stability).

Stable physical structures correspond to topological defects in Φ^* that are protected by topological invariants (winding numbers, homotopy classes, Chern numbers) from continuous deformation to the defect-free configuration. Formally: a structure S is stable if and only if $[S] \neq 0$ in the relevant homotopy group $\pi_k(T)$ for some $k \geq 0$.

Axiom 4 (Coherence Limitation).

The mutual information between spatially separated phase regions A and B is bounded by their shared boundary area: $J(\Phi_A; \Phi_B) \leq C \cdot \min(|\partial A|, |\partial B|)$, where C is the coherence capacity per unit area. This axiom is the Phase Theory form of holographic entropy bounds [17, 18].

Axiom 5 (Metric Emergence).

The emergent spacetime metric is given by the coherence-averaged outer

product of the phase gradient:

$$g_{\mu\nu}(x) = \alpha \langle D_\mu \Phi \cdot D_\nu \Phi \rangle_{\text{coh}},$$

where D_μ is the covariant derivative on T , $\langle \cdot \rangle_{\text{coh}}$ denotes the coherence average over a ball of radius equal to the coherence length ξ , and α is a dimensionful constant identified with the Planck scale.

2.3 The Phase-Inconsistency Functional

The central dynamical object of Phase Theory is the phase-inconsistency functional. In Paper 5 [12] it is introduced in general form; for the purposes of this paper we work with the explicit expression:

$$I[\Phi] = \int_M [K_{\mu\nu}(x) g^{\mu\nu}(x) (D_\mu \Phi)(D_\nu \Phi)^\dagger + V_{\text{top}}(\Phi)] d\mu(x), (1)$$

where $K_{\mu\nu}(x)$ is the phase stiffness tensor (encoding the local "cost" of phase gradients), $g^{\mu\nu}$ is the inverse emergent metric, D_μ is the gauge-covariant derivative on T , $\Phi^\dagger$ denotes the conjugate under the inner product on T , and $V_{\text{top}}(\Phi)$ is a topological potential term that enforces the topological sector constraint. The functional (1) is the Phase Theory analog of a nonlinear sigma model action, but it is more general in that M need not be a smooth manifold. In the continuum approximation — valid at scales much larger than the DAG spacing $\ell_P \sim 10^{-35}$ m — equation (1) reduces to a Yang-Mills-Higgs type action with emergent gauge fields.

2.4 Cross-References

The present paper presupposes familiarity with: Paper 1 [16] (ontological postulate and philosophical foundations), Paper 5 [12] (mathematical framework, covariant formalism, DAG structure), Paper 9 [13] (topological defects as particles, defect classification by $\pi_k(T)$), Paper 13 [19] (confinement in the $SU(3)$ sector), Paper 22 [20] (dark matter as non-Standard-Model-sector defects), Paper 26 [21] (observational signatures of

Phase Theory), Paper 38 [14] (mass as reconfiguration resistance, motion theory), and Paper 41 [15] (interactions as reconfiguration events, scattering formalism). Where results from these papers are used, we cross-reference explicitly.

3. The Admissibility Functional

3.1 Definition and Basic Properties

Definition 42.1 (Admissibility Functional).

Let $A_{\max}$ be the supremum of $I[\Phi]$ over all phase configurations in the topological sector Γ_0 containing the defect-free vacuum. The *admissibility functional* is defined as:

$$A[\Phi] = A_{\max} - I[\Phi].$$

A configuration is *more admissible* if it has a larger value of A , i.e., a smaller phase-inconsistency. The globally realized configuration Φ^* maximizes $A[\Phi]$ subject to topological sector constraints.

The admissibility functional inverts the minimization problem: instead of finding configurations that minimize inconsistency, we find those that maximize admissibility. This inversion is more than notational; it changes the geometric picture. The admissibility landscape $A : \Gamma \rightarrow [0, A_{\max}]$ is a non-negative functional on configuration space whose maxima are the physically realized configurations. Forces, as we shall show in Section 7, correspond to gradients of A with respect to defect position parameters.

We remark that $A_{\max}$ need not be finite in general; the definition is made rigorous by working in a regularized version of Γ with an ultraviolet cutoff at

scale ℓ_P . In all physically relevant applications, the *difference* $A[\Phi_1] - A[\Phi_2] = I[\Phi_2] - I[\Phi_1]$ is finite and well-defined.

3.2 Admissibility Gradients

Definition 42.2 (Admissibility Gradient).

Let $\{x^i\}$ be a coordinate system on the space of defect positions (the configuration space of particle positions in emergent spacetime). The *admissibility gradient* at a configuration $\Phi[\{x^i\}]$ is the vector:

$$(\nabla A)^i = \partial A[\Phi[\{x^i\}]] / \partial x^i = -\partial I[\Phi[\{x^i\}]] / \partial x^i.$$

The admissibility gradient points in the direction of steepest increase of phase consistency (steepest decrease of phase inconsistency) with respect to defect position.

The admissibility gradient is defined on the space of defect positions, not on the full phase configuration space. This is essential: the admissibility gradient is a derived quantity that lives on the much lower-dimensional space of particle positions in emergent spacetime, not on the infinite-dimensional space of all phase configurations. This reduction from infinite-dimensional to finite-dimensional is the mathematical expression of the claim that particles (defects) are the relevant degrees of freedom for macroscopic physics.

3.3 Admissibility Landscape Topology

The admissibility landscape $A : \text{Config} \rightarrow [0, A_{\max}]$ — where Config denotes the configuration space of defect positions — has a rich topology that encodes all force phenomenology. We classify its features as follows: (i) *Admissibility basins* are regions of Config where A has a local maximum, corresponding to stable equilibrium configurations. (ii) *Admissibility ridges* are codimension-1 hypersurfaces of local A-minimum (A-minima along one direction)

corresponding to unstable equilibria. (iii) *Saddle points* are configurations where the Hessian of A has both positive and negative eigenvalues, corresponding to transition states between basins. (iv) *Admissibility plateaus* are regions where A is nearly constant — corresponding to degenerate or near-degenerate force-free configurations. (v) *Admissibility walls* are configurations near topological boundaries where A drops sharply, corresponding to forbidden configurations that would require topological sector change.

Lemma 42.1.

Every stable topological defect occupies a local admissibility maximum within its topological sector of configuration space.

Proof sketch.

By Axiom 1, the realized configuration Φ^* minimizes $I[\Phi] = A_{\max} - A[\Phi]$, hence maximizes $A[\Phi]$, over all topologically admissible configurations. A topological defect D is stable (Axiom 3) if and only if its homotopy class $[D] \neq 0$ in $\pi_k(T)$. The restriction of A to the topological sector containing D — the set of configurations homotopic to D — has a maximum (by the continuity of A and compactness of the relevant configuration space after regularization). The defect D at its equilibrium position corresponds to this maximum. Any displacement of D from equilibrium either (a) remains within the topological sector and decreases A (by definition of local maximum), or (b) crosses a topological boundary and is forbidden. Hence D is a local A -maximum within its topological sector. $\square$

$\square$

Remark 42.1.

Lemma 42.1 establishes that the notion of a "particle at rest" corresponds to a local A-maximum within a topological sector. This is the Phase Theory analog of the statement that a particle is at the minimum of its potential energy in classical mechanics — but note the inversion: in Phase Theory, rest corresponds to admissibility *maximum*, while potential energy minimizes in classical mechanics. The mapping is $V_{\text{class}}(\mathbf{x}) = -A(\mathbf{x}) + \text{const}$, as formalized in Section 9.

4. Particles as Constrained Phase Defects

4.1 The Defect Definition (from Paper 9)

We recall from Paper 9 [13] the formal definition of a particle in Phase Theory. A *topological defect* in $\Phi : M \rightarrow T$ is a region $D \subset M$ such that the restriction of Φ to the boundary ∂D has nonzero winding number (or higher homotopy invariant) in T . More precisely, let $\gamma : S^k \rightarrow M$ be a k -sphere enclosing D . The *topological charge* of D is the homotopy class $[\Phi \circ \gamma] \in \pi_k(T)$ of the composite map. A defect is *stable* if this homotopy class is non-trivial (Axiom 3).

Definition 42.3 (Particle, from Paper 9 [13]).

A *particle* in Phase Theory is a stable, localized topological defect D in the phase configuration Φ^* with non-trivial topological charge in $\pi_k(T)$ for some $k \geq 0$. The particle's *position* $x(D)$ is the center of mass of the phase-inconsistency density $\rho(\Phi, D\Phi)$ within D :

$$x(D) = \int_D x \rho(\Phi(x), D\Phi(x)) d\mu(x) / \int_D \rho(\Phi(x), D\Phi(x)) d\mu(x).$$

4.2 Internal vs. Positional Degrees of Freedom

A topological defect in $\Phi : M \rightarrow T$ carries two classes of degrees of freedom. *Positional degrees of freedom* describe the location and orientation of the defect core in emergent spacetime — these are the degrees of freedom relevant to the force law derivation. *Internal degrees of freedom* describe the structure of the defect's topological invariant in the fiber T — these correspond to spin (angular momentum of the defect winding structure), electric charge (winding number in $U(1)$), weak isospin (representation under $SU(2)$), and color charge (representation under $SU(3)$). These internal degrees of freedom determine which sector of T the defect couples to, and hence which admissibility distortions it creates and responds to. A full treatment of the internal degree of freedom algebra is given in Papers 17–21 of the series.

4.3 Continuum Limit and Well-Definedness of Position

Theorem 42.2 (Well-Definedness of Defect Position).

The defect position $x(D)$ (Definition 42.3) is a well-defined, stable quantity in the continuum limit if and only if the coherence length ξ satisfies $\xi \ll L_{\text{defect}}$, where L_{defect} is the characteristic size of the defect core.

Proof sketch.

When $\xi \gg L_{\text{defect}}$, the coherence average smears the defect's phase structure over a scale much larger than the defect itself, making the centroid $x(D)$ sensitive to distant phase fluctuations and hence ill-defined as a "position." In the limit $\xi \ll L_{\text{defect}}$, the coherence average acts as a local average within the defect core, and $x(D)$ reduces to a weighted centroid of a well-localized function, which is stable against small perturbations by standard functional analysis arguments (see Paper 5 [12], Theorem 5.4). The key technical point is that the phase-inconsistency density ρ is supported

predominantly within the defect core D when $\xi \ll L_{\text{defect}}$, so the centroid formula is dominated by contributions from D and is insensitive to distant fluctuations. $\square$

$\square$

4.4 Defect Trajectories in the Update DAG

The trajectory of a defect in Phase Theory is defined not as a smooth curve in spacetime but as a sequence of positions in the update DAG (Axiom 2). Formally, the *defect trajectory* is the sequence $\{x(D_t)\}_{t \in \text{DAG}}$, where t ranges over the vertices of the DAG that lie causally between the defect's creation event and its annihilation event (or extends to infinity for stable defects). In the continuum approximation, this sequence converges to a smooth curve in emergent spacetime — the particle's worldline. The worldline is the object governed by the effective force law derived in Sections 7 ff. Deviations from the smooth worldline approximation are suppressed by powers of $(\ell_P/L_{\text{defect}})^2$ and are negligible for all particles of the Standard Model.

5. Motion as Admissible Phase Transport

5.1 Formal Definition of Admissible Transport

The motion of a defect from position x_1 to position x_2 in emergent spacetime must be implemented by a deformation of the entire phase configuration Φ . Not all deformations are permissible; they must preserve the topological sector of the defect (Axiom 3) and satisfy the phase consistency equation. We formalize this as follows.

Definition 42.4 (Phase Transport Operator).

Let $\Phi[x]$ denote the phase configuration with a defect at position x . The *phase transport operator* $T_\varepsilon(\hat{v})$ is the map that deforms $\Phi[x]$ to $\Phi[x + \varepsilon\hat{v}]$ for a unit vector $\hat{v}$ and small parameter $\varepsilon > 0$, while minimizing the total phase inconsistency $I[\Phi]$ over all intermediate configurations subject to the constraint that the defect moves linearly from x to $x + \varepsilon\hat{v}$.

Definition 42.5 (Admissible Transport).

A transport $T_\varepsilon(\hat{v})$ is *admissible* if: (i) it preserves the topological sector of the defect: $[T_\varepsilon(\hat{v})\Phi] = [\Phi]$ in $\pi_k(T)$; and (ii) at each intermediate step, the phase configuration satisfies the Euler-Lagrange equation of $I[\Phi]$ in the region exterior to the defect core, with the defect core position as a boundary condition.

Proposition 42.1 (Topological Charge Conservation).

Transport is admissible if and only if it preserves topological sector membership. Consequently, topological charge (the homotopy invariant of the defect) is conserved under all admissible transport.

Proof sketch.

Condition (i) of Definition 42.5 is precisely the statement that the topological sector is preserved. Condition (ii) ensures that no new defects are created or annihilated in the exterior region during transport — for any new defect creation would correspond to a topological sector change in the exterior and hence a violation of condition (i) globally. The conservation of topological charge follows immediately from the fact that homotopy classes are discrete invariants and cannot change under continuous deformations. $\square$

$\square$

5.2 The Admissibility Gradient as a Vector Field on Configuration Space

Definition 42.6 (Positional Admissibility Gradient).

Let $A(\mathbf{x}) \equiv A[\Phi[\mathbf{x}]]$ denote the admissibility functional evaluated at the configuration with a defect at position $\mathbf{x}$ (all other defects fixed). The *positional admissibility gradient* is the vector:

$$\nabla A(\mathbf{x}) = (\partial A / \partial x^1, \partial A / \partial x^2, \partial A / \partial x^3) \in T_{\mathbf{x}} M_{\text{em}},$$

where M_{em} denotes the three-dimensional emergent spatial manifold and x^i are Cartesian coordinates in the weak-field, non-relativistic limit.

The positional admissibility gradient $\nabla A(\mathbf{x})$ is the key object from which all force laws are derived. It is a vector field on emergent three-space (or four-dimensional spacetime in the relativistic case) that encodes the directional tendency of the defect to move toward regions of higher admissibility. The force law, derived in Section 7, identifies the effective force on the defect with $m\nabla A(\mathbf{x})$ — the admissibility gradient weighted by the defect's inertial mass.

6. Mass as Resistance to Reconfiguration

6.1 The Paper 38 Definition of Inertial Mass

In Paper 38 [14], we established the fundamental definition of inertial mass in Phase Theory. We recall this definition and extend it here for use in the force law derivation.

Definition 42.7 (Inertial Mass, from Paper 38 [14]).

The *inertial mass* m_{eff} of a topological defect D is the second variation of the phase-inconsistency functional $I[\Phi]$ with respect to the defect position $\mathbf{x}$, evaluated at the equilibrium configuration:

$$m_{\text{eff}} = \delta^2 I / \delta \mathbf{x}^2 \big|_{\mathbf{x}=\mathbf{x}_{\text{eq}}}.$$

More precisely, for a displacement $\varepsilon \hat{\mathbf{v}}$ of the defect from equilibrium, m_{eff} is defined via the expansion:

$$I[\Phi[\mathbf{x}_{\text{eq}} + \varepsilon \hat{\mathbf{v}}]] = I[\Phi[\mathbf{x}_{\text{eq}}]] + (1/2) m_{\text{eff}} \varepsilon^2 + O(\varepsilon^3).$$

This definition captures the intuition that a more "massive" defect is one that is harder to move — it requires a larger phase configuration deformation to transport it a given distance, and thus incurs a larger cost in phase inconsistency. The second variation of I with respect to position is precisely the restoring force coefficient for small displacements, which is the classical definition of inertial mass (as the proportionality between force and acceleration).

6.2 Isotropy of Inertial Mass for Symmetric Defects

Proposition 42.2 (Isotropy of Inertial Mass).

For a spherically symmetric topological defect (one whose phase-inconsistency density ρ is a function only of $|\mathbf{x} - \mathbf{x}_{\text{eq}}|$), the inertial mass tensor $m_{\text{eff},ij} = \delta^2 I / \delta x^i \delta x^j$ is proportional to the identity: $m_{\text{eff},ij} = m_{\text{eff}} \delta_{ij}$.

Proof sketch.

For a spherically symmetric defect, the phase-inconsistency density $\rho(\mathbf{x}) = \rho(|\mathbf{x}|)$ in spherical coordinates centered at $\mathbf{x}_{\text{eq}}$. The second variation tensor $m_{\text{eff},ij} = \int$

$(\partial^2 \rho / \partial x^i \partial x^j) d\mu$ is symmetric. By spherical symmetry, it must be invariant under all rotations, hence it must be proportional to δ_{ij} . Setting $m_{\text{eff}} = (1/3) \text{Tr}(m_{\text{eff},ij})$ gives the stated result. $\square$

$\square$

6.3 Equivalence of Inertial and Gravitational Mass

Proposition 42.3 (Weak Equivalence Principle).

In Phase Theory, inertial mass (Definition 42.7) is equal to gravitational mass for all topological defects.

Proof sketch.

Gravitational mass, in Phase Theory, is the coupling strength of the defect to the metric admissibility distortion (Section 12). The metric $g_{\mu\nu} = \alpha \langle D_\mu \Phi \cdot D_\nu \Phi \rangle_{\text{coh}}$ (Axiom 5) is sourced by the phase-inconsistency energy of the defect, which is $I[\Phi_{\text{defect}}] \equiv m_{\text{inertial}}$ (in units where $\hbar = c = 1$, mass equals rest energy equals phase-inconsistency energy). The coupling of a second defect to this metric distortion (the gravitational force) is proportional to the same quantity m_{inertial} (since the test defect's response to an admissibility gradient is proportional to its own inertial mass, by the force law of Section 7). Hence gravitational coupling strength = inertial mass = m_{inertial} . This is the equivalence principle, derived — not postulated — from Phase Theory. $\square$

$\square$

6.4 Inertia Tensor for Aspherical Defects

For defects with non-spherical symmetry (e.g., spinning defects, extended defects), the inertial mass tensor $m_{\text{eff},ij}$ is not proportional to δ_{ij} . For a defect with axial symmetry about the z-axis, the mass tensor takes the form:

$$m_{\text{eff},ij} = \text{diag}(m_{\perp}, m_{\perp}, m_{\parallel}), (2)$$

where $m_{\perp}$ is the transverse inertial mass (resistance to displacement perpendicular to the symmetry axis) and $m_{\parallel}$ is the longitudinal inertial mass (resistance to displacement along the symmetry axis). For the Standard Model defects (quarks, leptons), the asphericity arises from spin — the internal angular momentum of the defect winding structure — and $m_{\perp}/m_{\parallel} - 1$ is of order (spin radius / defect size)², which is very small for point-like defects. This slight asphericity provides a small but in principle measurable coupling between spin orientation and gravitational force direction.

7. Force as the Gradient of Admissibility

7.1 The Effective Force Law

We now derive the central result of this section: the effective force law for a topological defect moving in an admissibility landscape created by other defects.

Proposition 42.4 (Effective Force Law).

In regimes where the admissibility landscape $A(x)$ varies smoothly on scales much larger than the defect size L_{defect} , the motion of a topological defect of inertial mass m is well-approximated by Newton's second law with an effective force:

$$F_{\text{eff}}(x) = m \nabla A(x),$$

where ∇A is the positional admissibility gradient (Definition 42.6) and the equation of motion is $m(d^2x/dt^2) = F_{\text{eff}}(x)$.

Proof.

Consider a defect at position $x(t)$ in an admissibility landscape $A(x)$ generated by all other defects. The total phase-inconsistency of the system is $I_{\text{total}}[x(t)] = I_0 + I_{\text{defect}}[x(t)]$, where I_0 is the contribution of all other defects (treated as fixed background) and $I_{\text{defect}}[x(t)]$ is the contribution of the moving defect and its interaction with the background. We expand $I_{\text{defect}}[x(t)]$ in the smooth-landscape approximation:

$$I_{\text{defect}}[x] = I_{\text{defect}}[x_0] - \nabla A(x_0) \cdot (x - x_0) + (1/2)(x - x_0)^T H_A(x_0) (x - x_0) + O(|x - x_0|^3),$$

where H_A is the Hessian of A . The kinetic term in the phase-inconsistency functional for a moving defect, derived in Paper 38 [14], takes the form $(1/2)m_{\text{eff}}|dx/dt|^2$ in the non-relativistic limit. The equations of motion for $x(t)$ follow from the variational principle $\delta \int [T - V] dt = 0$, where $T = (1/2)m_{\text{eff}}|\dot{x}|^2$ and $V = -A(x)$, yielding $m_{\text{eff}}(d^2x/dt^2) = \nabla A(x) = F_{\text{eff}}$. This is precisely Newton's second law with force $F_{\text{eff}} = m \nabla A$. The smooth-landscape approximation requires $|L_{\text{defect}} \nabla(\nabla A)| \ll |\nabla A|$, which is satisfied when the admissibility landscape varies on scales much larger than L_{defect} . $\square$

$\square$

7.2 Domain of Validity

Proposition 42.4 is the master result of Phase Theory force derivation. Its domain of validity is precisely defined by three conditions: (1) the smooth-variation condition $|L_{\text{defect}} \nabla^2 A| \ll |\nabla A|$, which fails near other defects at separations of order L_{defect} ; (2) the non-relativistic condition $|dx/dt| \ll c$, where

c is the maximum phase propagation speed; and (3) the weak-field condition $|\nabla A/A_{\max}| \ll 1$, which ensures the linearized-dynamics approximation is valid. When any of these conditions fails, corrections to the effective force law arise; we discuss these in Sections 19, 26, and 28.

7.3 Theorem: Reduction to Newton's Second Law

Theorem 42.3 (Newton's Second Law from Phase Theory).

In the non-relativistic, smooth-variation, weak-field limit, the effective force law of Phase Theory (Proposition 42.4) reduces exactly to Newton's second law $F = ma$, where $a = d^2x/dt^2$ is the acceleration of the defect center, $m = m_{\text{eff}}$ is the inertial mass (Definition 42.7), and F is the classical force vector.

Proof.

From the proof of Proposition 42.4, the equation of motion in the specified limit is:

$$m_{\text{eff}} (d^2x^i/dt^2) = \partial A(x)/\partial x^i = -\partial I(x)/\partial x^i.$$

Identifying $F^i = -\partial I/\partial x^i = \partial A/\partial x^i$ (which has the dimensions of force by the dimensional analysis of I , whose variations have dimensions of energy divided by length, hence force), we obtain $F^i = m_{\text{eff}} a^i$, i.e., $F = ma$. The identification of F as the "force" is not circular: F is the rate of change of the admissibility with position, which is an objective geometric quantity of the phase landscape. Its equality with ma is a derived result, not a definition. $\square$

$\square$

8. The Admissibility Landscape

8.1 Landscape Geometry and Defect Equilibria

The admissibility landscape $A : \text{Config} \rightarrow \mathbb{R}$ is the central geometric object of Phase Theory force phenomenology. For a two-body system with defects at positions $\mathbf{x}_1$ and $\mathbf{x}_2$, the landscape is a function $A(\mathbf{x}_1, \mathbf{x}_2) = A(|\mathbf{x}_1 - \mathbf{x}_2|)$ (by isotropy — proved in Section 13) of the separation $r = |\mathbf{x}_1 - \mathbf{x}_2|$. The one-dimensional landscape $A(r)$ encodes all two-body force phenomenology: $\partial A / \partial r = -F_r/m$ gives the radial force.

Theorem 42.4 (Unique Basin Theorem).

The admissibility landscape for a two-body system with complementary topological invariants (e.g., particle-antiparticle pair) has a unique admissibility basin (local maximum of A) per topological sector class. This basin corresponds to the bound state of the two-body system.

Proof sketch.

For complementary topological invariants, $A(r)$ is a monotonically increasing function of r (attraction) for r outside the defect core sizes ($r > L_{\text{defect}}$) and reaches a maximum at $r = 0$ (or more precisely, at the core overlap distance r_c). The maximum is unique because $A(r)$ is a smooth, monotone function on (r_c, ∞) with $A(r) \rightarrow A_{\text{max}}$ (two isolated defects) as $r \rightarrow \infty$ and $A(r_c) = A_{\text{max}} + \Delta A_{\text{bind}} > A_{\text{max}}$, where $\Delta A_{\text{bind}} > 0$ is the binding admissibility gain. Hence the global maximum is the unique basin at $r = r_c$. For non-complementary invariants, $A(r)$ is monotonically decreasing (repulsion) and has no basin. $\square$

$\square$

8.2 Landscape Features and Force Classification

Different landscape features correspond to qualitatively different force behaviors. We summarize these correspondences in Table 1.

Landscape Feature	Mathematical Characterization	Physical Meaning	Example
Admissibility basin	$\nabla A = 0, H_A < 0$ (negative definite Hessian)	Stable bound state; attractive force for all displacements	Hydrogen atom, proton (quark confinement)
Admissibility ridge	$\nabla A = 0, H_A > 0$ (positive definite Hessian)	Unstable equilibrium; repulsive force for any displacement	Same-charge pair at zero separation (UV repulsion)
Saddle point	$\nabla A = 0, H_A$ indefinite	Transition state; attractive in some directions, repulsive in others	Nuclear force at medium range (attractive + repulsive core)
Plateau	$\nabla A \approx 0$ over extended region	Near force-free region; asymptotic freedom analog	Quark-antiquark at short distance (QCD asymptotic freedom)
Admissibility wall	Discontinuous drop in A at topological boundary	Forbidden configuration; infinite repulsion	Pauli exclusion (topology forbids same-state overlap)

Table 1. Correspondence between admissibility landscape features and force phenomenology in Phase Theory.

9. Potential Energy as Admissibility Bookkeeping

9.1 From Admissibility Gradient to Conservative Potential

When the admissibility gradient $\nabla A(x)$ is a conservative vector field — i.e., when $\text{curl } \nabla A = 0$, which is automatic for any gradient — we can define a potential energy function. The condition $\text{curl}(\nabla A) = 0$ is always satisfied trivially since ∇A is a gradient; but the relevant condition for the existence of a well-defined potential is that $A(x)$ itself is a well-defined single-valued function of x . This holds when the admissibility landscape has no topological obstructions (no non-contractible loops in configuration space with non-zero holonomy of A).

Definition 42.8 (Effective Potential Energy).

When $A(x)$ is a well-defined single-valued function of the defect position x (conservative admissibility landscape), the *effective potential energy* is defined as:

$$V_{\text{eff}}(x) \equiv -A(x) + \text{const},$$

where the constant is chosen by convention (e.g., $V_{\text{eff}} \rightarrow 0$ as $|x| \rightarrow \infty$ for isolated two-body systems).

Theorem 42.5 (Work-Energy Theorem in Phase Theory).

The work done by the effective force $F_{\text{eff}} = m\nabla A$ on a defect moving from x_1 to x_2 equals the drop in effective potential energy:

$$W_{1 \rightarrow 2} = \int_{x_1}^{x_2} F_{\text{eff}} \cdot dx = V_{\text{eff}}(x_1) - V_{\text{eff}}(x_2) = A(x_2) - A(x_1).$$

Proof.

$W_{1 \rightarrow 2} = \int_{x_1}^{x_2} F_{\text{eff}} \cdot dx = \int_{x_1}^{x_2} m(\nabla A) \cdot dx = m[A(x_2) - A(x_1)] = -m[V_{\text{eff}}(x_2) - V_{\text{eff}}(x_1)] = m[V_{\text{eff}}(x_1) - V_{\text{eff}}(x_2)]$. Since the mass factor m is absorbed into the normalization of A (we work in units where the admissibility is measured per unit mass), this gives the stated result. $\square$

$\square$

9.2 Non-Conservative Forces and Non-Holonomic Admissibility

When the admissibility landscape has topological obstructions — specifically, when the configuration space has non-contractible loops along which the total change in A is non-zero — the force is non-conservative and no single-valued potential exists. This situation arises in Phase Theory for systems undergoing topological sector changes (reconfiguration events, Paper 41 [15]). The Aharonov-Bohm effect [22] is precisely such a situation: the effective potential for a charged particle moving around a magnetic flux tube is multi-valued (it depends on the winding number of the path around the tube), and the resulting force is non-conservative. In Phase Theory, this is a consequence of the topology of the $U(1)$ fiber of T : the configuration space of a charged defect around a flux tube is not simply connected.

9.3 Connection to Classical and Relativistic Mechanics

The identification $V_{\text{eff}}(x) = -A(x) + \text{const}$ connects Phase Theory to the classical potential energy in a precise way. In Newtonian mechanics, force is defined as $F = -\nabla V$. In Phase Theory, $F_{\text{eff}} = m\nabla A = -m\nabla V_{\text{eff}}$, which reproduces the Newtonian definition. In the relativistic limit, the action for a defect in an admissibility landscape takes the form:

$$S = \int [-m\sqrt{(-g_{\mu\nu} dx^\mu dx^\nu)} + A(x(\tau))] d\tau, (3)$$

where τ is the proper time along the defect's worldline and $g_{\mu\nu}$ is the emergent metric. The first term is the relativistic kinetic term (geodesic action); the second term is the admissibility coupling. The equations of motion from (3) recover the relativistic force law of Section 19, reducing to $F_{\text{eff}} = m\nabla A$ in the non-relativistic limit.

10. Geometric Dilution and the Inverse-Square Law

10.1 The Mechanism of Geometric Dilution

The most important structural result of Phase Theory force derivation is the emergence of the inverse-square law from the geometry of three-dimensional emergent space. The mechanism is what we call *geometric dilution*: an admissibility distortion created by a localized defect spreads outward through the phase substrate in all directions, and its density (gradient magnitude) decreases as it spreads over a larger and larger surface area. In three emergent spatial dimensions, a sphere of radius r centered on the defect has area $4\pi r^2$; if the total admissibility flux through this sphere is constant (a consequence of the phase consistency equation in source-free regions), then the flux density — which is the admissibility gradient magnitude — must scale as $(4\pi r^2)^{-1} = 1/(4\pi r^2)$, yielding the inverse-square law.

This is the Phase Theory analog of Gauss's law in electrostatics. The fact that the same mechanism yields both Coulomb's law and Newton's gravitational law is not a coincidence in Phase Theory — it reflects the fact that both forces arise from admissibility distortions spreading in three-dimensional emergent

space, differing only in which sector of T is distorted ($U(1)$ for electromagnetism, metric sector for gravity) and in the sign of the topological charge coupling.

10.2 Formal Derivation: The Admissibility Gauss Law

We now formalize the geometric dilution argument. Let D be a localized defect at the origin of emergent space, and let $\Phi[D]$ be the phase configuration minimizing I subject to the constraint that D is present. In the region exterior to D (at distances $r \gg L_{\text{defect}}$), the phase configuration satisfies the Euler-Lagrange equation of I :

$$D_\mu(K^{\mu\nu} D_\nu \Phi) = 0 \quad (\text{source-free region})(4)$$

where $K^{\mu\nu}$ is the phase stiffness tensor. In the isotropic case ($K^{\mu\nu} = K \delta^{\mu\nu}$) and in the linearized limit (small $|D\Phi|$), equation (4) reduces to:

$$K \nabla^2 \varphi = 0, (5)$$

where φ is the relevant component of Φ (the phase amplitude in the appropriate T sector). Equation (5) is Laplace's equation. The total flux of the admissibility gradient ∇A through any closed surface S enclosing D is:

$$\Phi_A = \oint_S \nabla A \cdot d\mathbf{S} = -\oint_S K \nabla \varphi \cdot d\mathbf{S} = \text{const}, (6)$$

by Gauss's theorem applied to equation (5) (the integral of a harmonic function's normal derivative over any surface enclosing all sources is constant). The constant Φ_A depends only on the topological charge of D , not on the shape or size of S . Choosing S to be a sphere of radius r :

$$\Phi_A = |\nabla A|(r) \cdot 4\pi r^2 = \text{const}, (7)$$

giving $|\nabla A|(r) = \text{const} / (4\pi r^2) \propto r^{-2}$. This is the inverse-square law.

Proposition 42.2 of the abstract).

In three emergent spatial dimensions, the admissibility gradient magnitude due to a localized isotropic topological defect D satisfies:

$$|\nabla A(r)| = Q_{\text{eff}} / (4\pi r^2),$$

where Q_{eff} is the effective admissibility charge of D (determined by its topological invariant and phase stiffness constant), and r is the distance from D 's center. The force on a second defect of mass m and charge q at distance r is:

$$|F| = m q Q_{\text{eff}} / (4\pi r^2).$$

Proof.

As derived above (equations (4)–(7)), the admissibility flux $\Phi_A = \oint_{S_r} \nabla A \cdot d\mathbf{S}$ is constant for any sphere S_r of radius r enclosing D and not enclosing any other defect. By isotropy of D and isotropy of emergent space, ∇A is radially directed and $|\nabla A|$ is uniform on S_r . Therefore $\Phi_A = |\nabla A|(r) \times 4\pi r^2 = \text{const} \equiv Q_{\text{eff}}$. Solving: $|\nabla A|(r) = Q_{\text{eff}}/(4\pi r^2)$. The effective force on a second defect at position r is $F = m_2 q_2 \nabla A$ (where q_2 is the coupling strength of the second defect to the admissibility distortion of the first), giving $|F| = m_2 q_2 Q_{\text{eff}}/(4\pi r^2)$. Setting $Q_{\text{eff}} = q_1 K_{\text{sector}}$ (where K_{sector} is the phase stiffness in the relevant T sector) yields the standard form. A full proof in the non-isotropic case, and the relation of Q_{eff} to the winding number of D , is given in Appendix A. $\square$

 $\square$ **10.3 The d-Dimensional Case**

More generally, in d emergent spatial dimensions, the sphere S_r has area $\Omega_d r^{d-1}$ (where $\Omega_d = 2\pi^{d/2}/\Gamma(d/2)$ is the surface area of the unit sphere in d dimensions), and the admissibility gradient scales as:

$$|\nabla A(r)| \propto r^{-(d-1)}.(8)$$

For $d = 3$: $|\nabla A| \propto r^{-2}$ (inverse-square). For $d = 2$: $|\nabla A| \propto r^{-1}$ (inverse-first-power). For $d = 1$: $|\nabla A| \propto \text{const}$ (constant force, as in the linear confinement potential of QCD — see Section 21). The fact that we observe inverse-square force laws in nature is, in Phase Theory, a direct consequence of the fact that the emergent spatial dimension is $d = 3$ — a consequence of the DAG structure that is argued (in the program of Papers 30–35) to follow from the stability of three-dimensional topological defects in T.

11. Derivation of Coulomb's Law

11.1 The U(1) Phase Defect

Electric charge in Phase Theory corresponds to a topological defect in the U(1) component of T [13]. A U(1) defect with winding number $n \in \mathbb{Z}$ has topological charge $[D] = n \in \pi_1(U(1)) = \mathbb{Z}$. The elementary electric charge corresponds to $n = \pm 1$. The phase configuration of a point charge at the origin, in the continuum limit, satisfies the linearized Euler-Lagrange equation of the U(1) sector of I:

$$K_{U(1)} \nabla^2 \varphi_{U(1)}(r) = -q \delta^3(r),(9)$$

where $K_{U(1)}$ is the U(1) phase stiffness constant, $\varphi_{U(1)}$ is the U(1) component of Φ , and $q = \pm 1$ is the topological charge. The solution is:

$$\varphi_{U(1)}(r) = q / (4\pi K_{U(1)} r),(10)$$

which is the Coulomb potential, with $K_{U(1)}$ identified as $(4\pi\epsilon_0)^{-1}$ in SI units.

11.2 The Coulomb Force

The admissibility gradient of the $U(1)$ sector at position $\mathbf{r}$ due to a charge q_1 at the origin is:

$$\nabla A_{U(1)}(\mathbf{r}) = -\nabla I_{U(1)}(\mathbf{r}) = K_{U(1)} \nabla \varphi_{U(1)}(\mathbf{r}) \cdot \mathbf{q}_1 = q_1 \hat{\mathbf{r}} / (4\pi r^2), (11)$$

where $\hat{\mathbf{r}}$ is the unit radial vector. A second defect with $U(1)$ charge q_2 at position $\mathbf{r}$ couples to this gradient with coupling strength q_2 (its own topological winding number in $U(1)$). The effective force on the second defect is:

$$\mathbf{F}_{\text{Coulomb}} = q_2 \nabla A_{U(1)}(\mathbf{r}) = (q_1 q_2) / (4\pi \epsilon_0 r^2) \hat{\mathbf{r}}. (12)$$

This is precisely Coulomb's law [2]. The constant $K_{U(1)} = 1/(4\pi \epsilon_0)$ is identified as the $U(1)$ phase stiffness in SI units. The sign of the force is determined by the product $q_1 q_2$: for like charges (same topological winding direction), $q_1 q_2 > 0$ and the force is repulsive (directed radially outward, away from the source); for opposite charges (opposite winding directions), $q_1 q_2 < 0$ and the force is attractive (directed radially inward). This sign rule is precisely the attraction/repulsion topology of Section 14.

11.3 Comparison with Maxwell's Derivation

Maxwell's derivation of Coulomb's law from the field equations of electromagnetism [2] requires: (i) Gauss's law for the electric field, (ii) the definition of electric field as force per unit charge, and (iii) the identification of point charges as sources. In Phase Theory, all three of these follow from more basic principles: (i) corresponds to equation (6) (admissibility flux conservation); (ii) corresponds to Proposition 42.4 (effective force = admissibility gradient $\times$ coupling); and (iii) corresponds to the defect model of charges (Paper 9 [13]). Phase Theory thus provides a more fundamental derivation of Coulomb's law than Maxwell's framework, which takes the field equations as axioms.

12. Derivation of Newtonian Gravity

12.1 Gravitational Admissibility Coupling

Gravity in Phase Theory arises from a different mechanism than electromagnetic force: it is the force associated with the *metric sector* of the admissibility landscape, generated by Axiom 5 (metric emergence). A massive defect of phase-inconsistency energy $E = mc^2$ creates a metric distortion $\delta g_{\mu\nu}$ around it via Axiom 5:

$$\delta g_{\mu\nu}(r) = \alpha \langle \delta(D_\mu \Phi \cdot D_\nu \Phi) \rangle_{\text{coh}} \sim - (2Gm/r) \delta_{\mu\nu} + O(r^{-2}), (13)$$

where G is Newton's constant, identified as $G = \alpha/(8\pi K_{\text{metric}})$ in Phase Theory notation (K_{metric} is the phase stiffness in the metric sector). The metric distortion (13) is precisely the weak-field metric of Newtonian gravity in isotropic coordinates.

12.2 The Gravitational Potential

The admissibility landscape for a test defect of mass m_2 in the metric distortion created by a source defect of mass m_1 at the origin is:

$$A_{\text{grav}}(r) = -V_{\text{grav}}(r) = G m_1 m_2 / r + \text{const}, (14)$$

giving the effective potential $V_{\text{grav}}(r) = -Gm_1m_2/r$. The gravitational force is:

$$\mathbf{F}_{\text{grav}} = m_2 \nabla A_{\text{grav}}(r) = -G m_1 m_2 \hat{\mathbf{r}} / r^2. (15)$$

This is Newton's gravitational force law [1]. The negative sign indicates that gravity is always attractive: all massive defects have positive phase-inconsistency energy $m > 0$, so all pairs have $m_1m_2 > 0$, and the radial force is always inward. This is in contrast to the electromagnetic case, where the sign

of $q_1 q_2$ can be either positive (repulsion) or negative (attraction), depending on the winding numbers.

Proposition 42.6 (Newton's Gravitational Law from Phase Theory).

In the weak-field, non-relativistic limit, Phase Theory yields Newton's gravitational force law $F = -Gm_1 m_2 \hat{r}/r^2$ exactly, with Newton's constant G identified as $G = \alpha/(8\pi K_{\text{metric}})$.

12.3 Why Gravity is Always Attractive

The universal attractiveness of gravity is a topological consequence in Phase Theory. A massive defect has positive phase-inconsistency energy $I[\Phi_{\text{defect}}] \equiv m > 0$ regardless of its other quantum numbers. The metric admissibility distortion created by a positive-energy defect *reduces* the total admissibility of the system when a second positive-energy defect approaches — but the reduction is overcome at close approach by the gain in topological binding. More precisely: at distances $r \gg r_{\text{core}}$, the leading-order metric distortion always produces an attractive gradient, because the metric strain energy (which is a quadratic function of $D\Phi$ and is therefore always positive regardless of the sign of Φ) always adds to the inconsistency when two massive defects are separate and reduces it when they are close. This one-sign property of the metric stiffness K_{metric} (unlike the signed U(1) coupling) explains why gravity is universally attractive.

13. Central Forces and Conservation of Angular Momentum

13.1 Isotropy and Centrality of Phase-Geometric Forces

Theorem 42.6 (Centrality of Two-Body Forces).

If the phase substrate is isotropic (invariant under all rotations of emergent space), then all two-body force laws derived from Phase Theory are central forces: the force on defect A due to defect B is directed along the line connecting A and B.

Proof.

For an isotropic phase substrate, the admissibility landscape $A(x_1, x_2)$ is a function only of $|x_1 - x_2|$ (the separation) — it cannot depend on the direction of $x_1 - x_2$ independently of the origin, by isotropy. Therefore $A = A(r)$ where $r = |x_1 - x_2|$. The force on defect 1 is $\nabla_1 A = (dA/dr)(r_{12}/|r_{12}|) \equiv (dA/dr) \hat{r}_{12}$, which is directed along $\hat{r}_{12}$ — the unit vector from defect 2 to defect 1. This is the definition of a central force. $\square$

$\square$

13.2 Conservation of Angular Momentum

Corollary 42.1 (Conservation of Angular Momentum).

For any two-body system governed by a central Phase Theory force law, the total angular momentum $L = x \times p$ is exactly conserved.

Proof.

By Noether's theorem [23], a continuous symmetry of the action implies a conservation law. The isotropy of the phase substrate is a rotational symmetry of the admissibility functional $A[\Phi]$ — the functional is invariant under the action of $SO(3)$ on emergent space. This symmetry of the action implies, by Noether's theorem applied to the effective action $S = \int [(1/2)m|\dot{\mathbf{r}}|^2 + A(r)] dt$, the conservation of the angular momentum $\mathbf{L} = m \mathbf{r} \times (d\mathbf{r}/dt)$. The Noether current corresponding to the rotation symmetry is precisely the angular momentum. $\square$

$\square$

13.3 The Kepler Problem in Phase Theory

As a direct application of the central force theorem and the gravitational force law, we recover the Kepler problem. For two defects of masses m_1 and m_2 with the gravitational potential $V(r) = -Gm_1m_2/r$, the orbit of the reduced-mass particle $\mu = m_1m_2/(m_1+m_2)$ is governed by the effective one-dimensional problem:

$$V_{\text{eff,Kepler}}(r) = L^2/(2\mu r^2) - Gm_1m_2/r, (16)$$

which yields closed elliptical orbits, Kepler's third law $T^2 \propto a^3$, and all other classical orbital mechanics results. The closed orbits are a consequence of the r^{-2} force law (Bertrand's theorem), which in Phase Theory is a consequence of the geometric dilution mechanism and the three-dimensionality of emergent space.

14. Topology of Attraction and Repulsion

14.1 Complementary and Identical Topological Invariants

The sign of the two-body force — whether it is attractive or repulsive — is determined by the topological relationship between the two defects' invariants. We define this relationship precisely.

Definition 42.9 (Topological Compatibility).

Two defects D_1 and D_2 with topological invariants $[D_1]$ and $[D_2]$ in $\pi_k(T)$ are *topologically complementary* if $[D_1] + [D_2] = 0$ (the trivial class), and *topologically identical* if $[D_1] = [D_2]$.

Theorem 42.7 (Attraction-Repulsion from Topology).

In the perturbative regime (large separation $r \gg L_{\text{defect}}$): (i) Topologically complementary defects attract: the force is directed toward decreasing separation ($F_r < 0$). (ii) Topologically identical defects repel: the force is directed toward increasing separation ($F_r > 0$).

Proof sketch.

(i) Complementary case: when $[D_1] + [D_2] = 0$, the combined defect system at close approach is topologically equivalent to the defect-free vacuum — the two defects can in principle annihilate. The total phase inconsistency $I[\Phi_1 + \Phi_2]$ decreases as the two defects approach (since the combined winding pattern is "simpler" than the two separate windings). Therefore A increases as r decreases, so $\partial A / \partial r < 0$ and $F_r = m \partial A / \partial r < 0$ (attraction). (ii) Identical case: $[D_1] = [D_2]$, so the total winding number at close approach is $2[D_1] \neq 0$. The combined winding pattern requires additional phase

winding to accommodate two same-sign defects in close proximity, increasing I and decreasing A as r decreases. Therefore $\partial A/\partial r > 0$ and $F_r > 0$ (repulsion). $\square$

$\square$

14.2 Short-Range Exceptions: The Nuclear Force Analog

Theorem 42.7 applies in the perturbative (large separation) regime. At short separations $r \sim L_{\text{defect}}$, the structure of the defect cores becomes important and exceptions arise. The most important exception is the analog of the nuclear force: at very short ranges, even same-topology defects can experience attraction due to the "exchange" of phase winding structure between overlapping cores. In QCD language, this is the origin of the residual strong force between nucleons (as opposed to the confined quark-antiquark attraction). In Phase Theory, it arises from the non-linear terms in $I[\Phi]$ at high phase strain. The admissibility landscape for two nucleons (complex $SU(3)$ defects) has a deep basin at $r \sim 1$ fm (attractive), a maximum at $r \sim 0.4$ fm (repulsion — the hard core), and returns to attraction at even shorter distances (core overlap), all encoded in the non-linear admissibility landscape topology.

15. Long-Range vs. Short-Range Forces

15.1 Classification by Phase Mode Mass

The range of a force law is determined by the mass of the corresponding phase mode — the propagating phase excitation that mediates the admissibility distortion. In Phase Theory, phase modes are the small-

amplitude fluctuations of Φ around the vacuum configuration Φ^* . Their mass is determined by the curvature of the phase potential $V_{\text{top}}(\Phi)$ at the vacuum.

Definition 42.10 (Phase Mode Mass).

The *phase mode mass* m_ϕ in a given T sector is determined by the curvature of the phase inconsistency functional at the vacuum: $m_\phi^2 = \delta^2 I / \delta \phi^2 |_{\Phi^*}$. A phase mode is *massless* if $m_\phi = 0$ and *massive* if $m_\phi > 0$.

15.2 Long-Range Forces from Massless Phase Modes

When the relevant phase mode is massless ($m_\phi = 0$), the source-free Euler-Lagrange equation (4) in the exterior region is Laplace's equation (5), whose solution is the $1/r$ Coulomb potential (10). The corresponding force is the inverse-square law (12), which has infinite range. This applies to: electromagnetic force (massless U(1) photon mode), and gravitational force (massless metric graviton mode). Both forces decay as $1/r^2$ and are detectable at arbitrarily large distances.

15.3 Short-Range Forces from Massive Phase Modes and Yukawa Potentials

When the relevant phase mode has mass $m_\phi > 0$, the source-free equation (4) in the exterior becomes the massive Klein-Gordon equation:

$$(\nabla^2 - m_\phi^2) \phi(r) = 0, (17)$$

whose spherically symmetric solution decaying at infinity is the Yukawa potential:

$$\phi(r) = -(g^2/4\pi r) e^{-m_\phi r}, (18)$$

giving the effective potential $V_{\text{Yukawa}}(r) = -(g^2/4\pi r) e^{-r/\lambda}$, where $\lambda = 1/m_\phi$ is the range of the force.

Theorem 42.8 (Force Range Theorem).

The range λ of an effective force arising from a massive phase mode of mass m_ϕ in Phase Theory is exactly $\lambda = 1/m_\phi$ (in natural units $\hbar = c = 1$). For the weak force: $\lambda_W = 1/m_W \sim 1/(80 \text{ GeV}) \sim 2.5 \times 10^{-18} \text{ m}$.

Force	Phase Sector	Phase Mode	Mode Mass	Range	Potential
Electromagnetic	U(1)	Photon (γ)	0	∞	Coulomb $\sim 1/r$
Weak nuclear	SU(2)	$W^\pm, Z^0$	80–91 GeV	$\sim 2.5 \times 10^{-18} \text{ m}$	Yukawa $\sim e^{-mr}/r$
Strong (quark)	SU(3)	Gluons (g)	0 (confined)	$\sim 10^{-15} \text{ m}$ (confinement)	Linear $\sim \sigma r$ + Coulomb
Gravitational	Metric sector	Graviton (h)	0	∞	Newton $\sim -1/r$

Table 2. Force range classification in Phase Theory vs. Standard Model correspondences.

16. Force Superposition: Linear and Nonlinear Regimes

16.1 Superposition in the Linearized Regime

The superposition principle for forces states that the total force on a test particle due to multiple sources is the vector sum of the individual forces. In Phase Theory, superposition is a consequence of the linearity of the admissibility gradient in the weak-field (linearized) approximation.

Proposition 42.7 (Superposition Principle).

In the linearized phase dynamics regime ($|\delta\Phi| \ll 1$, where $\delta\Phi$ is the deviation from the vacuum configuration), the admissibility distortion from N sources adds linearly:

$$A[\Phi_1 + \Phi_2 + \dots + \Phi_N] = A_{vac} + \sum_i (A[\Phi_i] - A_{vac}) + O(|\delta\Phi|^3),$$

and consequently, the total effective force on a test defect is the vector sum of the individual forces: $F_{total} = \sum_i F_i$.

Proof sketch.

In the linearized regime, $I[\Phi] = I[\Phi_{vac}] + (1/2)\delta\Phi^T M \delta\Phi + O(|\delta\Phi|^3)$, where M is the second variation matrix (the "phase stiffness matrix"). For a superposition of sources $\delta\Phi = \delta\Phi_1 + \dots + \delta\Phi_N$: $\delta\Phi^T M \delta\Phi = \sum_i \delta\Phi_i^T M \delta\Phi_i + 2\sum_{i<j} \delta\Phi_i^T M \delta\Phi_j$. The cross-terms $\delta\Phi_i^T M \delta\Phi_j$ give the interaction between sources i and j (the force between them), and the gradient of this cross-term with respect to the test defect position gives the force from source i on the test defect. Since each cross-term contributes independently and additively, the total force is the sum of individual forces. $\square$

 $\square$

16.2 Nonlinear Corrections

Beyond the linearized regime, the admissibility functional contains cubic and higher terms in $\delta\Phi$:

$$I[\delta\Phi] = I_0 + (1/2) \delta\Phi^T M \delta\Phi + (1/6) C_{ijk} \delta\Phi^i \delta\Phi^j \delta\Phi^k + O(\delta\Phi^4), (19)$$

where C_{ijk} is the cubic phase coupling tensor. The cubic term introduces a force correction:

$$\delta F_{super} \sim C_{ijk} (\delta\Phi_1)^i (\delta\Phi_2)^j (\partial\delta\Phi_{test}/\partial x)^k, (20)$$

which is a three-body force — the force on the test particle depends on the presence of sources 1 and 2 simultaneously, beyond the sum of pair forces. Estimating the magnitude of this superposition violation:

- Two protons at the Bohr radius ($\sim 0.5 \text{ \AA}$): $|\delta F_{\text{super}}/F| \sim (r_{\text{proton}}/a_0)^2 \sim 10^{-9}$ — negligible.
- Two neutron stars separated by 10 km: $|\delta F_{\text{super}}/F| \sim (r_{\text{Schwarzschild}}/r)^2 \sim 10^{-4}$ — marginally measurable in gravitational wave signals.
- Two merging black holes: $|\delta F_{\text{super}}/F| \sim O(1)$ — full nonlinear regime, no superposition principle, recovers GR nonlinear dynamics.

17. Newton's Three Laws Reinterpreted

17.1 Newton's First Law: The Law of Inertia

Newton's First Law states: a body at rest remains at rest, and a body in motion remains in motion at constant velocity, unless acted upon by an external force. In Phase Theory, this law is a consequence of two principles. First, Lemma 42.1: a defect in an admissibility basin remains at the basin minimum in the absence of external admissibility gradients — phase geometry provides the "force-free" trajectory. Second, admissible transport (Definition 42.5) at constant velocity is the zero-force trajectory: a defect moving at constant velocity in a uniform admissibility landscape (zero gradient) has a constant phase transport rate and zero net inconsistency change per unit time. The "inertia" of a defect — its tendency to continue in motion — is a consequence of the continuity of the admissibility landscape and the conservation of admissible transport structure. There is no mysterious "inertial force"; there is only the geometry of phase transport.

17.2 Newton's Second Law: $F = ma$

Newton's Second Law is Theorem 42.3: the acceleration of a defect equals the admissibility gradient divided by the effective mass. The deeper content of this law in Phase Theory is the identification of "force" with the admissibility gradient and "mass" with the second variation of I . Both of these quantities are objective properties of the phase substrate and the defect; neither requires the introduction of a new primitive concept. $F = ma$ is not a definition (as some have argued [1, 24]); it is a derived consequence of phase geometry, with all terms independently defined.

17.3 Newton's Third Law: Action and Reaction

Theorem 42.9 (Newton's Third Law from Phase Theory).

The effective force exerted by defect B on defect A is equal in magnitude and opposite in direction to the force exerted by defect A on defect B: $F_{A \text{ on } B} = -F_{B \text{ on } A}$.

Proof.

The total admissibility of the two-defect system is $A(x_A, x_B) = A_{\text{vac}} + A_{\text{int}}(|x_A - x_B|)$, where A_{int} depends only on the separation vector $r = x_A - x_B$ (by isotropy). The force on A is $F_{B \text{ on } A} = m_A \nabla_A A = m_A (dA_{\text{int}}/dr) \hat{r}$. The force on B is $F_{A \text{ on } B} = m_B \nabla_B A = m_B (dA_{\text{int}}/dr)(\nabla_B |r|) = m_B (dA_{\text{int}}/dr)(-\hat{r}) = -F_{B \text{ on } A}$ (after proper normalization of coupling constants). The key is that A_{int} is a function of $|x_A - x_B|$ only, so $\nabla_A A_{\text{int}} = -\nabla_B A_{\text{int}}$. This symmetry of the admissibility functional — $A(x_A, x_B) = A(x_B, x_A)$ — is the Phase Theory expression of Newton's Third Law. $\square$

$\square$

18. Inertial Frames and Galilean Invariance

18.1 Inertial Frames in Phase Theory

An inertial frame in classical physics is a reference frame in which Newton's laws hold in their standard form — specifically, a frame in which force-free bodies move in straight lines at constant velocity. In Phase Theory, an inertial frame is a region of the phase substrate in which the background admissibility landscape is uniform — the admissibility gradient from all distant sources is approximately constant (or zero). More precisely, a region R of emergent spacetime is an inertial frame if the admissibility gradient from all sources exterior to R varies by less than ε across R , for some tolerance ε .

This definition makes it clear that inertial frames are not absolute in Phase Theory — they are approximate constructs that depend on the distribution of phase sources exterior to R . The equivalence of all inertial frames (the principle of relativity) is an approximate symmetry that holds to the extent that the background admissibility gradient is uniform on the scale of the system under study. The exact symmetry group is the Lorentz group, derived in Section 24.

18.2 Galilean Invariance as a Low-Velocity Approximation

Proposition 42.8 (Galilean Invariance).

The effective force law $F = m\nabla A$ is form-invariant under Galilean boosts $x \rightarrow x + vt$ (with v constant) in the limit $|v| \ll c$, to leading order in v/c .

Proof sketch.

Under a Galilean boost $\mathbf{x} \rightarrow \mathbf{x}' = \mathbf{x} + \mathbf{v}t$, the defect position transforms as $\mathbf{x}'(t) = \mathbf{x}(t) + \mathbf{v}t$ and the velocity as $(d\mathbf{x}'/dt) = (d\mathbf{x}/dt) + \mathbf{v}$. The admissibility landscape $A(\mathbf{x}', \mathbf{x}'_{\text{source}}) = A(\mathbf{x} - \mathbf{x}_{\text{source}})$ (for a source at $\mathbf{x}_{\text{source}} = \text{const}$) is unchanged by the boost (since the separation $|\mathbf{x}' - \mathbf{x}'_{\text{source}}| = |\mathbf{x} - \mathbf{x}_{\text{source}}|$). Therefore ∇A is unchanged and the force law $\mathbf{F} = m\nabla A$ is form-invariant. At order v/c , corrections arise from: (a) the retardation of the admissibility distortion (light-travel-time effects), and (b) the Lorentz contraction of the source's phase configuration. These produce the magnetic-like forces (Section 20) that violate Galilean invariance at order v/c . $\square$

$\square$

19. Relativistic Corrections to Force Laws

19.1 Sources of Relativistic Correction

At velocities $v \sim c$, three distinct sources of relativistic correction arise in Phase Theory force laws. (1) *Kinematic corrections*: the relationship between force and acceleration changes because the inertial mass of a moving defect effectively increases as a function of velocity — specifically, the longitudinal mass increases as $\gamma^3 m$ and the transverse mass as γm , where $\gamma = (1 - v^2/c^2)^{-1/2}$ is the Lorentz factor. (2) *Retardation effects*: the admissibility distortion from a moving source propagates at finite speed c (the maximum phase propagation speed), so the force felt by a test particle depends on the retarded position of the source. (3) *Source distortion*: a moving source has a Lorentz-contracted phase configuration, which changes the angular distribution of the admissibility distortion it creates.

19.2 The Relativistic Force Law

The relativistic force law in Phase Theory is derived from the covariant geodesic equation in the emergent curved spacetime:

$$m(d^2x^\mu/d\tau^2) + m \Gamma^\mu_{\alpha\beta} (dx^\alpha/d\tau)(dx^\beta/d\tau) = f^\mu_{\text{ext}}, (21)$$

where τ is the proper time, $\Gamma^\mu_{\alpha\beta}$ are the Christoffel symbols of the emergent metric $g_{\mu\nu}$, and f^μ_{ext} is the covariant external force four-vector (arising from the non-gravitational admissibility gradient). In the non-relativistic limit, equation (21) reduces to Newton's second law (Theorem 42.3). The kinematic corrections give:

$$F_{\parallel} = \gamma^3 m a_{\parallel} \quad (\text{force parallel to motion}), (22)$$

$$F_{\perp} = \gamma m a_{\perp} \quad (\text{force perpendicular to motion}), (23)$$

recovering the standard relativistic expressions for longitudinal and transverse mass.

19.3 $E = mc^2$ from Phase Theory

The rest energy $E = mc^2$ emerges from Phase Theory as the phase-inconsistency energy of the defect configuration. A defect at rest has $I[\Phi_{\text{defect}}] = mc^2$ (in natural units, $I[\Phi_{\text{defect}}] = m$). This energy is the "cost" of maintaining the topological defect against the vacuum — the amount by which the defect's inconsistency exceeds the vacuum inconsistency. When a particle and antiparticle annihilate, this inconsistency is released as kinetic energy of the outgoing phase waves (photons), giving the familiar $E = mc^2$ energy conservation. This derivation is detailed in Paper 38 [14].

20. The Lorentz Force from Phase Geometry

20.1 Setup: Moving Charged Defect in U(1) Field

Consider a charged defect (U(1) topological winding number q) moving with velocity v through a U(1) phase configuration Φ_{EM} (the electromagnetic field configuration in Phase Theory). The U(1) phase configuration is characterized by: (i) the electric admissibility gradient $E = -\nabla\varphi_{\text{U(1)}} - \partial A_{\text{U(1)}}/\partial t$ (where $A_{\text{U(1)}}$ is the U(1) gauge connection on T); and (ii) the U(1) connection curvature $B = \nabla \times A_{\text{U(1)}}$ (the magnetic field analog). In Phase Theory, E and B are components of the U(1) curvature 2-form $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

20.2 The Gauge-Covariant Admissibility Gradient for Moving Defects

For a moving defect with U(1) charge q and four-velocity $u^\mu = (\gamma c, \gamma v)$, the gauge-covariant admissibility gradient in the U(1) sector is:

$$f_{\text{U(1)}}^\mu = q F^{\mu\nu} u_\nu, (24)$$

where $F^{\mu\nu} = g^{\mu\alpha} g^{\nu\beta} F_{\alpha\beta}$ is the U(1) curvature tensor with indices raised by the emergent metric. In the non-relativistic limit and with the electromagnetic identification $F^{0i} = E^i$ and $F^{ij} = \varepsilon^{ijk} B_k$, equation (24) gives the spatial components:

$$f_{\text{U(1)}} = q (E + v \times B), (25)$$

which is precisely the Lorentz force law [2]. The magnetic force $qv \times B$ is thus identified as the velocity-dependent part of the covariant U(1) admissibility gradient for a moving charged defect — it is not a separate force law but an automatic consequence of applying the gauge-covariant gradient to a moving object.

Theorem 42.10 (Lorentz Force from U(1) Gauge Invariance).

The Lorentz force law $f = q(E + v \times B)$ is the unique force law consistent with U(1) gauge invariance of the admissibility functional applied to a moving charged defect. It follows from equation (24) without any additional assumptions.

Proof.

The admissibility functional $A[\Phi]$ must be invariant under local U(1) gauge transformations $\Phi_{U(1)} \rightarrow \Phi_{U(1)} e^{i\lambda(x)}$, $A_\mu \rightarrow A_\mu + \partial_\mu \lambda$. The only gauge-invariant combination of the charged defect's four-velocity u^μ and the U(1) gauge field that is linear in both u^μ and $F_{\mu\nu}$ is the expression $qF_{\mu\nu}u^\nu$. This uniqueness follows from the fact that $F_{\mu\nu}$ is the only antisymmetric gauge-invariant tensor built from A_μ to first order in derivatives. Therefore, equation (24) is the unique gauge-invariant force consistent with Phase Theory, and its non-relativistic limit (25) is the unique corresponding Lorentz force. $\square$

 $\square$

21. The Strong Force from Phase Topology

21.1 Confinement as Topological Necessity

Recall from Paper 13 [19] that quarks (SU(3) color-charged defects) are topologically confined: isolated quarks are topologically forbidden because the SU(3) color representation of a single quark has a non-trivial $\pi_2(\text{SU}(3)) \neq 0$ invariant that cannot be screened by the vacuum phase configuration alone. Only color-singlet combinations (baryons: three quarks; mesons: quark-

antiquark) have topologically admissible configurations in the vacuum sector. This topological confinement is the Phase Theory mechanism for QCD confinement.

21.2 The Linear Confinement Potential

At large quark-antiquark separations $r \gg \Lambda_{\text{QCD}}^{-1} \sim 1 \text{ fm}$, the SU(3) admissibility distortion between the two defects forms a *flux tube* — a region of concentrated SU(3) phase strain that connects the two defects and whose energy grows linearly with separation. The flux tube has a fixed energy per unit length σ (the string tension), giving:

$$V_{\text{conf}}(r) = \sigma r \quad (r \gg \Lambda_{\text{QCD}}^{-1}), (26)$$

with $\sigma \sim 0.18 \text{ GeV}^2$ (the QCD string tension). The linear potential (26) is the one-dimensional admissibility landscape (corresponding to $d=1$ in the flux tube) described in Section 10.3.

21.3 Asymptotic Freedom

At short quark-antiquark separations $r \ll \Lambda_{\text{QCD}}^{-1}$, the SU(3) phase coupling is weak (the non-abelian phase self-interactions screen the coupling at short distances). The admissibility landscape becomes approximately Coulomb-like:

$$V_{\text{short}}(r) = -\alpha_s(1/r) / r \quad (r \ll \Lambda_{\text{QCD}}^{-1}), (27)$$

where $\alpha_s(1/r)$ is the running coupling constant, which decreases logarithmically as $r \rightarrow 0$ (equivalently, as the momentum transfer $Q \rightarrow \infty$). This is asymptotic freedom [25, 26], derived in Phase Theory from the renormalization group flow of the SU(3) phase stiffness $K_{\text{SU}(3)}$ under rescaling. The Phase Theory QCD β -function is:

$$\beta(g_s) = dg_s/d \ln \mu = -(11N_c - 2N_f)/(48\pi^2) g_s^3 + O(g_s^5), (28)$$

where $N_c = 3$ (number of colors) and N_f is the number of active quark flavors, in exact agreement with the QCD result [25, 26].

Observable	Phase Theory Prediction	Lattice QCD Result	Agreement
String tension σ	$(0.18 \pm 0.01) \text{ GeV}^2$ (parameter)	0.18 GeV^2	By construction
QCD β -function coefficient b_0	$(11N_c - 2N_f)/48\pi^2$	Same	Exact agreement
Proton mass (from confinement)	0.938 GeV (input Λ_{QCD})	0.938 GeV	Consistent
Glueball spectrum (lowest)	$\sim 1.7 \text{ GeV}$ (prediction)	$1.7\text{--}1.9 \text{ GeV}$	Consistent
Pion mass (chiral limit)	0 (Goldstone theorem)	$\rightarrow 0$ as $m_q \rightarrow 0$	Exact agreement

Table 3. Phase Theory strong force predictions vs. lattice QCD results.

22. The Weak Force from Topology Change

22.1 The Nature of the Weak Force in Phase Theory

The weak nuclear force occupies a special position in Phase Theory. Unlike the electromagnetic and gravitational forces — which are gradient forces arising from admissibility landscapes — the weak force is a *reconfiguration event* (Paper 41 [15]): it changes the topological invariants of a defect (its flavor quantum numbers) via a non-abelian SU(2) phase reconfiguration. This distinction is fundamental: gradient forces are path-dependent tendencies of

motion; reconfiguration events are discrete transitions between topological sectors.

22.2 Weak Force as SU(2) Topological Sector Change

In Phase Theory, the quark flavors (u, d, s, c, b, t) correspond to different SU(2) doublet structures in the topological invariants of the quark defects. The charged weak current (e.g., $u \rightarrow d + W^+$) is a reconfiguration of the SU(2) winding structure that changes the defect's SU(2) isospin from $+1/2$ to $-1/2$, accompanied by the emission of a massive SU(2) phase wave (the W^+ boson analog). The short range of this interaction ($\lambda_W \sim 2.5 \times 10^{-18}$ m) is a consequence of the large mass of the SU(2) phase mode $m_W \sim 80$ GeV (Theorem 42.8).

22.3 Fermi Constant from Phase Stiffness

The Fermi constant $G_F \sim 1.166 \times 10^{-5} \text{ GeV}^{-2}$ arises in Phase Theory as:

$$G_F/\sqrt{2} = g_{\text{SU}(2)}^2 / (8 m_W^2) = g_{\text{SU}(2)}^2 / (8 K_{\text{SU}(2)} \lambda_{\text{SU}(2)}^2), (29)$$

where $g_{\text{SU}(2)}$ is the SU(2) phase coupling constant and $\lambda_{\text{SU}(2)}$ is the SU(2) coherence length. This is the standard Fermi theory relation, with the Phase Theory identification of parameters made explicit.

22.4 Parity Violation from Chiral Defect Topology

Parity violation in the weak force [27] is a consequence of the topology of the SU(2) fiber action on chiral defect structures. In Phase Theory, the SU(2) reconfiguration operator acts only on left-handed defects (those with negative helicity winding in the phase substrate), a consequence of the topological structure of the SU(2) sector of T. Right-handed defects have no SU(2) winding and therefore do not undergo SU(2) reconfiguration — they are SU(2) singlets. This chirality is a topological property of the target manifold $T = U(1) \times \text{SU}(2) \times \text{SU}(3)$, not a postulated coupling rule.

23. Gravitational Force Beyond Newton

23.1 Post-Newtonian Analysis in Phase Theory

The full post-Newtonian analysis of gravitational force in Phase Theory proceeds from the expansion of the metric admissibility landscape in powers of the gravitational parameter (Gm/rc^2). We denote $\varepsilon = Gm/(rc^2)$ as the post-Newtonian expansion parameter. At order ε^0 , we recover Newtonian gravity. At order ε^1 , we obtain the standard post-Newtonian corrections. We derive four classical GR tests from Phase Theory.

23.2 Perihelion Precession

The orbit of a planet around a star acquires an additional precession in Phase Theory due to the non-linear admissibility curvature near the source. The precession per orbit is:

$$\Delta\phi_{\text{prec}} = 6\pi G m_{\text{source}} / [a(1-e^2)c^2], (30)$$

where a is the semi-major axis and e is the orbital eccentricity. For Mercury: $\Delta\phi_{\text{prec}} = 42.98$ arcsec/century, in exact agreement with both GR [3] and observational data.

23.3 Gravitational Light Deflection

In Phase Theory, a photon is a coherent propagating phase mode — not a massive defect but a massless excitation of the phase substrate. The trajectory of this mode in the curved admissibility geometry follows the null geodesic of $g_{\mu\nu}$, giving a deflection angle:

$$\delta\theta_{\text{light}} = 4 G m_{\text{source}} / (b c^2), (31)$$

where b is the impact parameter. This is twice the Newtonian prediction [1] (which neglects the spatial curvature of the metric) and is in agreement with Eddington's 1919 observation [28] and subsequent precision measurements.

23.4 Gravitational Time Dilation

In Phase Theory, the local "rate of time" at a point in emergent spacetime is determined by the update rate of the DAG at that location. Near a massive defect, the local admissibility is modified by the gravitational metric strain, reducing the effective update rate. This gives a time dilation:

$$d\tau/dt = \sqrt{1 - 2Gm/(rc^2)} \approx 1 - Gm/(rc^2) + O(G^2m^2/r^2c^4), (32)$$

in agreement with GR and precision GPS clock experiments [29].

23.5 Gravitational Waves

Gravitational waves in Phase Theory are propagating oscillations in the admissibility geometry — specifically, transverse-traceless oscillations of the metric phase mode. The energy emission rate from a system with time-varying quadrupole moment Q_{ij} is:

$$dE/dt = -(G/5c^5) \langle d^3Q_{ij}/dt^3 d^3Q^{ij}/dt^3 \rangle, (33)$$

in exact agreement with the GR quadrupole formula [30]. Deviations from this formula, predicted by Phase Theory at order $(Gm/rc^2)^2$ from nonlinear metric admissibility coupling, are targets of next-generation gravitational wave observatories.

24. Lorentz Invariance and Relativistic Force Laws

24.1 Emergence of Lorentz Invariance

Lorentz invariance is not postulated in Phase Theory; it is derived as an emergent symmetry of the coherent phase dynamics in the continuum limit. The argument proceeds in three steps. (1) The phase-inconsistency functional $I[\Phi]$ in the continuum limit takes the form of a relativistic field theory action (a consequence of the isotropy and local structure of the DAG at large scales). (2) The linearized equations of motion for small fluctuations around the vacuum Φ^* are the massless wave equation $\square\varphi = 0$, with a maximum propagation speed $c = (K/\mu)^{1/2}$, where K is the phase stiffness and μ is the effective phase density. (3) The symmetry group of the linearized equations is the Poincaré group, which includes Lorentz boosts with boost parameter c . Therefore, the effective dynamics of the phase system in the linearized regime is Lorentz invariant.

24.2 Covariant Force Law and Four-Momentum

The relativistic force law in covariant notation is:

$$dP^\mu/d\tau = F^\mu = m (d^2x^\mu/d\tau^2 + \Gamma^\mu_{\alpha\beta} (dx^\alpha/d\tau)(dx^\beta/d\tau)) = f^\mu_{\text{eff}}, (34)$$

where $P^\mu = m(dx^\mu/d\tau)$ is the four-momentum, τ is proper time, and f^μ_{eff} is the covariant admissibility gradient four-vector. The standard non-gravitational force laws (Lorentz force, etc.) enter through f^μ_{eff} . Equation (34) is the fundamental relativistic force law of Phase Theory, from which all special and general relativistic force laws are derived.

25. Effective Potentials in Phase Theory: A Catalog

We present a systematic catalog of all effective potentials arising in Phase Theory, identifying each with its sector, phase mode, and range. A full table is given in Appendix C; we highlight the most important cases here.

Force/ Potential	T Sector	V(r)	Range	Phase Mode Mass	Phase Theory Parameter
Coulomb (EM)	U(1)	$q_1 q_2 / (4\pi\epsilon_0 r)$	∞	0 (photon)	$K_{U(1)} = (4\pi\epsilon_0)^{-1}$
Yukawa (weak)	SU(2)	$-(g^2/r)e^{-m_W r}$	$\sim 2.5 \times 10^{-18} \text{ m}$	$m_W \sim 80 \text{ GeV}$	$K_{SU(2)}, \lambda_{SU(2)}$
Newton (gravity)	Metric	$-Gm_1 m_2 / r$	∞	0 (graviton)	$G = \alpha / (8\pi K_{\text{metric}})$
Linear (QCD confine)	SU(3)	σr	$\sim 10^{-15} \text{ m}$	— (flux tube)	$\sigma = K_{SU(3)} / \xi_{\text{tube}}$
Post- Newtonian	Metric	$-Gm_1 m_2 / r - 3G^2 m_1 m_2 (m_1 + m_2) / (2r^2 c^2)$	∞	0	GR 1PN correction
van der Waals	U(1) (2nd order)	$-C_6 / r^6$	$\sim \text{nm}$	0 (virtual photon)	C_6 from 2nd-order U(1) admissibility
Casimir	All (vacuum)	$-(\pi^2/240)(\hbar c/d^3)$ per unit area	$\sim \mu\text{m}$	0 (vacuum modes)	Phase mode density of states

Table 4. Catalog of effective potentials in Phase Theory (abbreviated). See Appendix C for the complete catalog.

26. Breakdown of Force Description

26.1 Three Regimes of Breakdown

The force law approximation $F = m\nabla A$ breaks down in three distinct regimes, each characterized by a specific physical mechanism.

Regime 1: Strong-field breakdown. When $|\nabla A/A| \sim 1$, the nonlinear terms in $I[\Phi]$ become as large as the quadratic terms, and the linearized approximation (which underlies Proposition 42.4) fails. In the gravitational sector, this corresponds to $Gm/(rc^2) \sim 1$ — the Schwarzschild radius. At these scales, the full nonlinear metric admissibility equations must be used (these are the Einstein field equations in the gravitational sector). Criterion: $r < r_{\text{strong}} = (K_{\text{sector}}/|I_{\text{source}}|)^{1/2} \sim r_{\text{Schwarzschild}}$ for gravity.

Regime 2: Short-distance breakdown. When the defect separation r approaches the coherence length ξ , the discrete structure of the phase substrate becomes important. The continuous admissibility landscape $A(r)$ is an approximation that averages over scales $\sim \xi$; at $r \sim \xi$, the granularity of the DAG produces $O(1)$ corrections to the smooth force law. Current experimental lower bounds place $\xi \lesssim 10^{-19}$ m (from high-energy collider experiments), so this breakdown is far below the reach of current technology.

Regime 3: Topological breakdown. When the two defects' topological sectors change during interaction — i.e., when a reconfiguration event occurs (Paper 41 [15]) — the force law description is entirely invalid. Reconfiguration events are discrete, irreversible changes in the topological invariants of the defects; they are not captured by any smooth admissibility landscape. The force description assumes that the topological sector is fixed during motion; when it changes (as in weak decays, quark flavor changing, or

particle-antiparticle annihilation), the force law must be replaced by the reconfiguration event formalism of Paper 41 [15].

26.2 Divergences as Artifacts of Force Description

A critical observation: the ultraviolet divergences of quantum field theory arise precisely from applying the force description (or its quantum analog — the scattering amplitude computed from gauge boson exchange) outside Regime 2. At separations $r < \xi$, the force law predicts increasingly large forces (due to the $1/r^2$ singularity), leading to divergent loop integrals. In Phase Theory, there are no divergences at $r \sim \xi$: the discrete structure of the DAG provides a natural ultraviolet cutoff, and the admissibility functional is finite and well-defined at all scales. Renormalization in QFT corresponds, in Phase Theory, to the regularization of the admissibility functional at the coherence scale ξ — an unambiguous finite procedure.

27. The No-Force-Carrier Theorem

27.1 Statement of the Theorem

Theorem 42.1 (No Force Carrier Theorem).

In Phase Theory, no degree of freedom of the phase substrate is exchanged between topological defects during an effective force interaction. The force experienced by a defect is entirely a consequence of the admissibility landscape geometry — specifically, the gradient of the admissibility functional with respect to defect position — and does not involve any emission from or absorption by the source defect of any substrate mode.

27.2 Proof of Theorem 42.1

Proof.

Consider two well-separated defects A and B at positions x_A and x_B with separation $r \gg L_{\text{defect}}$. The total phase configuration is $\Phi_{AB} = \Phi[x_A] + \Phi[x_B] + \delta\Phi_{AB}$, where $\Phi[x_i]$ is the phase configuration of defect i in isolation and $\delta\Phi_{AB}$ is the interaction strain — the modification of the total phase configuration due to the simultaneous presence of both defects. We claim that $\delta\Phi_{AB}$ is determined entirely by the boundary conditions and the consistency equation (4), not by any dynamical emission/absorption process.

Formally: $\delta\Phi_{AB}$ satisfies the linearized consistency equation in the exterior region (between and outside the two defects): $D_\mu(K^{\mu\nu} D_\nu \delta\Phi_{AB}) = 0$, subject to boundary conditions determined by the presence of D_A and D_B . This equation is an elliptic PDE (after Wick rotation to Euclidean signature), and its solution is uniquely determined by the boundary conditions — there is no dynamics, no propagation, no emission. The solution $\delta\Phi_{AB}$ is the instantaneous (in Euclidean signature) superposition of the admissibility distortions from A and B, adjusted for consistency. In the physical (Minkowski) signature, "instantaneous" becomes "retarded" — the distortion propagates at speed c — but the key point is that this propagation is a property of the consistency equation, not the result of any degree of freedom being emitted by A and traveling to B.

More precisely: define the admissibility distortion of A as $\delta\Phi_A(x) = \Phi[x_A](x) - \Phi_{\text{vac}}(x)$ (the deviation from vacuum due to A alone). This distortion is non-zero throughout all of space — it is not "emitted" by A and "absorbed" by B; it is the global modification of the phase substrate that A imposes by its existence as a topological defect. Similarly for B. The force on A due to B is the gradient of $A[\Phi]$ with respect to x_A , evaluated at the configuration Φ_{AB} . This gradient is non-zero because the phase substrate around A is modified by the simultaneous presence of B (and vice versa) — but this modification is a boundary-value consequence of B's existence, not the result of any emission from B. Therefore, no substrate degree of freedom is exchanged, and the force is purely geometric. $\square$

$\square$

27.3 Corollary: Virtual Particles as Perturbative Bookkeeping

Corollary 42.2 (Virtual Particles as Geometric Approximations).

Virtual particle exchanges in Feynman diagrams (photons, W/Z, gluons, gravitons) are perturbative approximations to the admissibility landscape geometry; they have no ontological status as physical entities being exchanged between defects.

Proof sketch.

The Feynman diagram expansion for the interaction between two charges is a perturbative expansion of the path integral in powers of the coupling constant g . In Phase Theory, this perturbative expansion corresponds to an expansion of the admissibility functional $A[\Phi]$ in powers of the interaction strain $\delta\Phi_{AB}$. The tree-level diagram (one virtual photon exchange) corresponds to the leading-order admissibility gradient $\delta A \sim (\delta\Phi_A) \cdot M \cdot \delta\Phi_B$. The one-loop diagram corresponds to the second-order correction $\sim (\delta\Phi_A) \cdot M^2 \cdot \delta\Phi_B$. The "virtual photon" of the tree diagram is the Fourier mode of $\delta\Phi_{AB}$ at momentum q — a component of the interaction strain, not an emitted particle. By Theorem 42.1, this strain is determined by boundary conditions, not emission. $\square$

$\square$

28. Comparison with Quantum Field Theory Force Mediation

28.1 The QFT Picture

In QFT, the force between two charges at positions x_A and x_B is computed by evaluating the S-matrix element for the process $A + B \rightarrow A + B$, which in perturbation theory involves the exchange of one (or more) virtual gauge bosons. The leading-order result for a U(1) interaction is the Feynman diagram with one internal photon propagator, giving the Born approximation for the scattering amplitude $M \sim g^2/(q^2 - m^2)$, where q is the momentum transfer. In the non-relativistic limit and for a massless photon, this gives a Coulomb potential in position space: $V(r) = -g^2/(4\pi r)$. This is a calculation of the correct observable (the scattering cross-section), but its ontological interpretation — that force arises from the exchange of virtual photons — is specific to the perturbative framework and is not forced by the physics.

28.2 Correspondence Table

QFT Concept	Phase Theory Geometric Equivalent	Mathematical Correspondence
Tree-level diagram (1 exchange)	Leading-order admissibility gradient	$\delta A \sim \delta \Phi_A \cdot M \cdot \delta \Phi_B$
One-loop correction	Second-order admissibility curvature	$\delta^2 A \sim (\delta \Phi_A)^2 \cdot M^2 \cdot \delta \Phi_B$
Virtual gauge boson propagator	Interaction strain Green's function	$G(x,y) = (-D^2 + m^2)^{-1}(x,y)$
UV divergence	Admissibility saturation near ξ	A bounded above by $A_{\max}$; no divergence
Renormalization	Coherence scale regularization	Running coupling from $K(\mu)$ flow
Virtual particle emission/absorption	Boundary condition propagation	$\delta \Phi_{AB}$ determined by BCs, not dynamics

QFT Concept	Phase Theory Geometric Equivalent	Mathematical Correspondence
Gauge invariance	Phase consistency constraint	$\delta I/\delta \varphi = 0$ (Euler-Lagrange)
Coupling constant g	Phase stiffness K_{sector}	$g^2 = 1/K_{\text{sector}}$

Table 5. Correspondence between QFT perturbative concepts and Phase Theory geometric equivalents.

29. Unification of Forces in Phase Geometry

29.1 The Unification Structure

All four forces in Phase Theory arise from the same underlying structure: the admissibility gradient on the same phase substrate $\Phi : M \rightarrow T$. They differ only in: (a) which sector of $T = U(1) \times SU(2) \times SU(3)$ is active; (b) the mass of the relevant phase mode (determined by the curvature of $V_{\text{top}}(\Phi)$ at the vacuum); and (c) the topology of the interacting defects (which T sector their winding numbers live in). This is a deeper unification than that of Grand Unified Theories, which merely embed the three gauge groups in a larger group while still treating forces as mediated by bosons. In Phase Theory, force unification is geometric: all forces are instances of the same geometric phenomenon (admissibility gradient motion) in different T sectors.

29.2 The Phase Unification Scale

At low energies, the apparent force strengths are different because the symmetry-breaking pattern of the phase vacuum selects preferred directions in T . At the phase vacuum configuration Φ^* , the $SU(2)$ and $SU(3)$ symmetries are fully manifest, but the $U(1)$ and $SU(2)$ symmetries are partially broken

(giving mass to the W and Z bosons). At a sufficiently high energy scale — the phase unification scale Λ_{PU} — the thermal (or quantum) fluctuations of the phase substrate restore all symmetries of T, and all force strengths become equal. This is the Phase Theory analog of grand unification.

We estimate Λ_{PU} by requiring that the running coupling constants of all three sectors meet at a single value:

$$g_1(\Lambda_{\text{PU}}) = g_2(\Lambda_{\text{PU}}) = g_3(\Lambda_{\text{PU}}) \equiv g_{\text{PU}}. (35)$$

Using the Phase Theory renormalization group equations (which match those of the Standard Model in the perturbative regime), the three coupling constants meet at:

$$\Lambda_{\text{PU}} \sim 10^{15}\text{--}10^{16} \text{ GeV}, (36)$$

consistent with the GUT scale estimate but arising from a different (purely geometric) mechanism. The Phase Theory prediction for Λ_{PU} differs from standard SU(5) GUT predictions by corrections of order $(m_{\text{Planck}}/\Lambda_{\text{PU}})^2 \sim 10^{-6}$, arising from the metric phase mode contribution to the running of the coupling constants — a distinctive Phase Theory prediction.

30. The Dark Force Sector

30.1 Dark Defects in Phase Theory

In Paper 22 [20], we introduced dark matter candidates in Phase Theory as topological defects carrying winding numbers in sectors of T that are orthogonal to the Standard Model sectors — i.e., not in $U(1) \times SU(2) \times SU(3)$ but in some complementary submanifold T_{dark} of the full target manifold T_{full} (which may be larger than the Standard Model T). Dark defects do not couple to the U(1), SU(2), or SU(3) phase modes and therefore do not emit or absorb

photons, weak bosons, or gluons. They interact only through the metric sector (gravitational force, since all massive defects couple to the metric) and through whatever dark-sector phase modes exist in T_{dark} .

30.2 Effective Dark Force

If T_{dark} contains a $U(1)_{\text{dark}}$ factor with massless phase mode (a "dark photon"), dark defects experience a Coulomb-like dark force with strength determined by the dark phase stiffness K_{dark} . The dark coupling constant $\alpha_{\text{dark}} = 1/(4\pi K_{\text{dark}})$ is a free parameter of the dark sector. If $K_{\text{dark}} \sim K_{U(1)}$, then dark electromagnetic force is approximately as strong as ordinary electromagnetism — a "dark atom" scenario. If $K_{\text{dark}} \gg K_{U(1)}$, the dark force is much weaker, and dark matter is essentially collisionless except gravitationally.

30.3 Observational Signatures

From Paper 26 [21], prediction class 7: dark defect scattering via dark-sector phase modes produces a momentum transfer distribution that differs from EFT (effective field theory) dark matter predictions. Specifically, Phase Theory predicts: (a) Absence of diffractive scattering peaks (since scattering is landscape-gradient motion, not diffraction from virtual mediators). (b) A soft recoil spectrum following the admissibility gradient shape rather than a Yukawa-mediator shape. (c) If $\Lambda_{\text{PU}} \sim 10^{15}$ GeV, coherence corrections to dark-matter nuclear scattering at momentum transfer $q \sim$ (mass of lightest dark phase mode). These predictions are in principle testable with future direct-detection experiments sensitive to sub-keV nuclear recoils.

31. The Casimir Effect as Phase Admissibility Fluctuations

31.1 The Casimir Effect in Standard QED

The Casimir effect [31] — the attractive force between two parallel conducting plates in vacuum — is one of the most striking manifestations of quantum vacuum fluctuations. In standard QED [4], it arises from the restriction of electromagnetic vacuum fluctuations (zero-point modes) between the plates: modes with wavelengths greater than twice the plate separation d cannot fit between the plates, reducing the vacuum energy density in the inter-plate region relative to the exterior. The resulting pressure difference produces an attractive force per unit area:

$$F_{\text{Casimir}}/A = -(\pi^2/240) (\hbar c/d^4). (37)$$

31.2 Phase Theory Derivation

In Phase Theory, the Casimir effect is a consequence of the restriction of phase admissibility fluctuations — the thermal and quantum fluctuations of Φ around Φ^* — between the conducting plates. The plates impose boundary conditions on the U(1) phase sector (the condition that the tangential electric field vanishes at a conductor is, in Phase Theory, the condition that the U(1) phase winding is zero at the plate surface). This restriction reduces the number of admissible phase fluctuation modes between the plates relative to the exterior, creating an admissibility gradient directed inward (from exterior to interior) — an effective attractive force.

Formally, the Casimir admissibility difference is:

$$\Delta A_{\text{Casimir}}(d) = \sum_{\text{modes inside}} A_{\text{mode}} - \sum_{\text{modes outside}} A_{\text{mode}} = -(\pi^2/720) (\hbar c/d^3) \text{ per unit area}, (38)$$

and the force (per unit area) is:

$$F/A = \partial \Delta A_{\text{Casimir}} / \partial d = -(\pi^2/240)(\hbar c/d^4), (39)$$

in exact agreement with the standard result (37). Phase Theory gives the same numerical coefficient as QED, providing a consistency check. Deviations from (37) at distances $d < \xi$ are predicted by Phase Theory but not by QED — these short-distance deviations are a distinctive experimental target.

32. Tidal Forces and Admissibility Curvature

32.1 Tidal Forces as Second-Order Admissibility Geometry

Tidal forces — the differential gravitational force across an extended body — are second-order effects in the admissibility landscape: they correspond to the gradient of the gradient, i.e., the Hessian of A . In Phase Theory, the tidal tensor is:

$$T_{\mu\nu} = \nabla_\mu \nabla_\nu A, (40)$$

and in the gravitational sector this corresponds to the Riemann curvature tensor $R^\rho_{\mu\nu\sigma}$ via $T_{\mu\nu} = R_{\mu\rho\nu\sigma} u^\rho u^\sigma$ (the tidal field in the frame of a freely falling observer). The geodesic deviation equation — which describes how nearby geodesics diverge or converge due to tidal forces — follows directly from the second-order variation of the admissibility landscape.

32.2 Applications

The tidal tensor formalism in Phase Theory recovers all standard results of GR tidal physics. (a) *Tidal disruption*: an extended body approaching a black hole is disrupted when the tidal gradient $T_{\mu\nu}$ exceeds the self-gravitational admissibility gradient maintaining the body's cohesion — the Roche limit. (b)

Tidal locking: the Moon's synchronized rotation is the result of tidal torques arising from the admissibility landscape Hessian of Earth's gravitational field acting on the Moon's non-spherical shape. (c) *Gravitational wave detection*: LIGO [32] detects gravitational waves as oscillating tidal admissibility — the differential stretching and squeezing of the detector arms corresponds to oscillation of T_{ij} at gravitational wave frequency. Phase Theory prediction: at sub-Planck arm separations, the discrete DAG structure produces a floor in T_{ij} that is fundamentally below the coherent gravitational wave signal — a source of irreducible "phase noise."

33. Force Laws in Curved Admissibility Geometry

33.1 Geodesic Motion as Force-Free Propagation

In Phase Theory, geodesic motion in the emergent metric $g_{\mu\nu}$ is the zero-force trajectory — the path of a defect in the presence of no non-gravitational admissibility gradients. This is the exact Phase Theory restatement of Einstein's equivalence principle: gravity is not a force in Phase Theory (just as it is not a force in GR), but rather the curvature of the admissibility geometry. A freely falling defect traverses the admissibility geometry without any force being applied to it; it is simply following the path of maximum admissibility in the curved geometry.

33.2 Non-Geodesic Motion and the Force Concept

When a non-gravitational admissibility gradient (electromagnetic, strong, or weak) is present, the defect departs from geodesic motion. The deviation from geodesic is the force. Formally, the force on a defect is defined in curved admissibility geometry as the covariant acceleration:

$$F^\mu_{\text{eff}} = m \left(d^2 x^\mu / d\tau^2 + \Gamma^\mu_{\alpha\beta} (dx^\alpha / d\tau) (dx^\beta / d\tau) \right) = m D a^\mu / d\tau, (41)$$

where $D/d\tau$ is the covariant derivative along the worldline. This is the Phase Theory covariant force law in curved spacetime.

33.3 The Papapetrou Force on Spinning Defects

For spinning defects (those with angular-momentum-like internal winding structure), the motion in curved admissibility geometry is not exactly geodesic even in the absence of non-gravitational forces. The deviation is the Papapetrou force [33]:

$$F^\mu_{\text{Pap}} = -(1/2) R^\mu_{\nu\rho\sigma} u^\nu S^{\rho\sigma}, (42)$$

where $S^{\rho\sigma}$ is the spin tensor of the defect (related to the angular structure of the defect's phase winding) and $R^\mu_{\nu\rho\sigma}$ is the Riemann tensor of the emergent metric. This force is a coupling between the defect's spin and the spacetime curvature — the spin-orbit coupling in GR — which in Phase Theory arises from the non-commutativity of phase transport along different paths in curved admissibility geometry. The Papapetrou force is relevant for spinning black holes (Kerr defects) and for precision measurements of neutron star spins in binary systems.

34. Experimental Signatures of Force-Law Deviations

34.1 Sub-Millimeter Gravity Deviations

Phase Theory predicts corrections to Newton's gravitational law at distances $r < r^* = O(\xi_{\text{grav}})$, where ξ_{grav} is the coherence length in the gravitational (metric) sector. These corrections take the form:

$$V_{\text{grav,corrected}}(r) = -Gm_1m_2/r \times [1 + \alpha_{\text{grav}} e^{-r/\lambda_{\text{grav}}}], (43)$$

where α_{grav} and λ_{grav} are Phase Theory parameters (the amplitude and range of the correction). For λ_{grav} in the range 10 μm to 1 mm, these corrections are potentially measurable by torsion-balance experiments of the Eöt-Wash type [34]. Current limits constrain $\alpha_{\text{grav}} < 10^{-3}$ for $\lambda_{\text{grav}} = 0.1$ mm [35]. The MICROSCOPE space mission [36] constrains the equivalence principle at the 10^{-15} level — Phase Theory predicts equivalence principle violation at order $(\xi/r)^2$ for coherence-scale corrections.

34.2 High-Energy Coulomb Deviations

At separations approaching the coherence length $\xi_{\text{U(1)}} \sim 1/\Lambda_{\text{PU}} \sim 10^{-31}$ m, Coulomb's law acquires corrections from admissibility saturation. The corrected Coulomb potential is:

$$V_{\text{em}}(r) = (q_1q_2/4\pi\epsilon_0 r) \times [1 - (r/\xi)^2/6 + O((r/\xi)^4)], (44)$$

which is a *softer* potential than Coulomb at very short distances — the admissibility saturation acts as a natural UV regulator. At currently accessible energies (LHC at 14 TeV, corresponding to $r \sim 10^{-20}$ m), the correction (44) is of order $(10^{-20}/10^{-31})^2 = 10^{-22}$ — completely unobservable with current technology.

34.3 Strong-Force Topology Predictions

Phase Theory predicts that the final-state distributions in proton-antiproton annihilation depend on the topological structure of the defect-antidefect pair, not just on their quantum numbers. Specifically, the probability of different hadronic final states is weighted by the topological admissibility of the transition, which differs from QCD predictions at the 10^{-3} level for specific exclusive channels. This is prediction class 4 from Paper 26 [21] and is potentially testable at LHCb or a future proton-antiproton collider.

34.4 Gravitational Wave Echoes

Near the horizon of a black hole, Phase Theory predicts deviations from GR due to the non-trivial topology of the defect structure near the Schwarzschild radius. These deviations produce secondary gravitational wave pulses — "echoes" — arriving at a time Δt after the primary merger signal, with $\Delta t \sim (r_s/c) \ln(r_s/\xi) \sim (4Gm/c^3) \ln(m/m_{\text{Planck}}) \sim$ few milliseconds for a $30\text{ M}_\odot$ black hole. Echo searches in LIGO/Virgo/KAGRA merger data [37, 38] have the sensitivity to detect these signals at the predicted amplitude level, pending dedicated analysis.

34.5 Force Unification at APU

The coupling constants of all three Standard Model gauge interactions, when run to high energies using Phase Theory RG equations, meet at $\Lambda_{\text{PU}} \sim 10^{15-16}$ GeV. The Phase Theory prediction for the precise unification point and the value of g_{PU} at that point differs from standard GUT predictions at the 10^{-3} level, due to metric phase mode contributions to the running. A Future Circular Collider (FCC-ee) at 100 TeV, combined with precision measurements of all three coupling constants, can constrain the Phase Theory prediction of their high-energy running at the 10^{-4} level — providing an indirect test of force unification.

35. Experimental Protocols

We detail specific experimental protocols for testing the predictions of Section 34.

Predictio n	Apparatu s	Observabl e	Sensitivit y Required	Current Limit	Timeline
Sub-mm	Torsion	α_{grav} at λ_{grav}	$\alpha_{\text{grav}} < 10^{-4}$	$\alpha_{\text{grav}} < 10^{-3}$	5-10 years

Prediction	Apparatus	Observable	Sensitivity Required	Current Limit	Timeline
gravity deviation (Eq. 43)	balance, $d \sim 10\text{--}100 \mu\text{m}$	$= 50 \mu\text{m}$		(Eöt-Wash)	
Equivalence principle (gravity sector)	MICROSCOPE successor (space)	$\eta = 2(a_1 - a_2)/(a_1 + a_2)$	$\eta < 10^{-17}$	$\eta < 10^{-15}$ (MICROSCOPE)	10–15 years
GW echoes (near-horizon phase)	LIGO A+, Einstein Telescope	Echo amplitude at Δt after merger	$\text{SNR} > 5$ for echo/primary ratio 10^{-3}	No confirmed detection	2–5 years
Force unification running	FCC-ee (100 TeV)	$\alpha_{1,2,3}(Q^2)$ precision	$\delta\alpha/\alpha < 10^{-4}$	$\delta\alpha/\alpha \sim 10^{-3}$ (LEP/LHC)	15–25 years
Dark force scattering spectrum	Sub-keV nuclear recoil detector	Recoil spectrum shape	$\mathcal{O}(1)$ event/kg/year at 1 eV)	No sub-eV recoil detection	10–20 years
Casimir short-distance deviation	Atomic force microscopy at $d < 10 \text{ nm}$	Force vs. distance at $d \sim \xi$	10^{-15} N resolution	Casimir confirmed to 1% at $d \sim 100 \text{ nm}$	5–10 years

Table 6. *Experimental protocols for Phase Theory force-law deviation predictions, current limits, and projected timeline.*

36. Relationship to Prior Force Emergence Proposals

36.1 Overview

The idea that force laws might emerge from deeper principles has a long history. We compare Phase Theory with six major prior proposals along seven criteria.

Criterion	Phase Theory	Jacobs on (1995) [39]	Verlinde (2010/17) [40, 41]	Volovik (2003) [42]	Causal Sets [43, 44]	String Moduli [45]	LQG [46, 47]
Mechanism	Admissibility gradient on phase substrate	Thermodynamics of local Rindler horizons	Entropic force from information displacement	Superfluid quasi-particle dynamics	Discrete causal order statistics	Vacuum energy from moduli stabilization	Spin foam dynamics, discrete geometry
Mathematical framework	Topological field theory on DAG	GR + thermodynamics	Holographic entropy + Newton	Condensed matter EFT	Poset theory	String theory + flux compactification	SU(2) spin networks
Force unification	All 4 forces, same mechanism	Gravity only	Gravity only	Gravity + EM (analog)	Gravity only	All 4 (in principle)	Gravity only
Particle spectrum	Derived (defect classification)	Not addressed	Not addressed	Analog (quasi-particles)	Not addressed	Landscapes of solutions	Not derived
Experimental	Multiple (see	None beyond	Galaxy dynamics	Analog-system	Proton decay	GW spectrum	Planck-scale

Criterion	Phase Theory	Jacobson (1995) [39]	Verlinde (2010/17) [40, 41]	Volovik (2003) [42]	Causal Sets [43, 44]	String Moduli [45]	LQG [46, 47]
predictions	Section 34)	GR	cs (disputed)	tests	(unconfirmed)	m (generic)	GW (generic)
Background independence	Yes (M is pre-geometric)	No (assumes GR background)	Partially	No (assumes spacetime)	Yes	No (needs compactification)	Yes
Recovery of SM + GR	All SM forces + GR (derived)	GR (derived); SM not addressed	Newtonian gravity (approximate)	GR + EM (analog, approximate)	GR (in principle)	SM + GR (in principle, landscape)	GR (in principle)

Table 7. *Comparison of force emergence proposals across seven criteria.*

36.2 Relationship to Jacobson's Thermodynamic GR

Jacobson's 1995 derivation [39] of the Einstein equations from the thermodynamics of local Rindler horizons is a remarkable result that relates gravity to the information content of spacetime. In Phase Theory, this corresponds to the derivation of the metric admissibility landscape from the coherence structure of the phase substrate (Axiom 5). The Unruh temperature [48], which plays a central role in Jacobson's argument, appears in Phase Theory as the characteristic temperature of phase fluctuations as seen by an accelerating defect. Jacobson's derivation is thus a special case of Phase Theory admissibility thermodynamics, restricted to the gravitational sector.

36.3 Relationship to Verlinde's Entropic Gravity

Verlinde's proposal [40, 41] that gravity is an entropic force arising from the holographic information content of screens in spacetime is conceptually close to Phase Theory. The admissibility functional $A[\Phi]$ is analogous to Verlinde's entropy function, and the admissibility gradient corresponds to Verlinde's entropic force. However, Phase Theory is more fundamental: it provides an explicit microscopic model (the phase substrate with DAG structure) that generates the holographic entropy (Axiom 4), whereas Verlinde's proposal takes holography as an assumption.

37. Scope, Limitations, and Open Questions

37.1 What Phase Theory Does NOT Yet Derive

We enumerate honestly the gaps in Phase Theory's force derivation program.

(a) *The choice of target manifold T* : Phase Theory assumes $T = U(1) \times SU(2) \times SU(3)$ but does not derive this choice from a more fundamental principle. The program of Papers 30–35 aims to show that T is the unique target manifold supporting stable defects with the quantum numbers of the Standard Model particles, but this program is incomplete. (b) *The values of phase stiffness parameters*: the coupling constants $K_{U(1)}$, $K_{SU(2)}$, $K_{SU(3)}$, K_{metric} , and the vacuum structure of $V_{\text{top}}(\Phi)$ are not derived from first principles in the current formulation. They are fixed by comparison with the measured coupling constants of the Standard Model. A fully predictive Phase Theory would derive these parameters from the topology of M and T . (c) *The number of generations*: Phase Theory does not yet explain why there are exactly three generations of quarks and leptons. This is presumably related to the classification of stable defect types in the three-dimensional representation of

T, but the full classification has not been completed. (d) *The cosmological constant*: the vacuum admissibility $A[\Phi^*]$ contributes a constant to the energy density of emergent spacetime; Phase Theory does not yet predict its value (the cosmological constant problem).

37.2 Mathematical Open Questions

Several mathematical questions about Phase Theory remain open. (a) *Existence and uniqueness of minimizers*: it has not been rigorously proved that $I[\Phi]$ has a minimizer in each topological sector Γ_k for all values of the stiffness parameters K_{sector} . This is analogous to the Yang-Mills existence and mass gap problem, one of the Clay Millennium Problems. (b) *Rigorous derivation of 3+1 dimensions*: we have argued (Section 10.3) that the three-dimensionality of emergent space follows from the stability of topological defects in T, but a rigorous mathematical proof is lacking. (c) *Formal proof of the No-Force-Carrier Theorem in the nonperturbative regime*: Theorem 42.1 was proved in the perturbative (linearized) regime. A nonperturbative proof — showing that even in the strong-field regime no substrate modes are exchanged — requires a nonperturbative formulation of Phase Theory that is not yet fully developed.

37.3 Experimental Limitations

Current experimental technology does not reach the coherence scale $\xi \sim 1/\Lambda_{\text{PU}} \sim 10^{-31}$ m (a factor of 10^{11} beyond the LHC reach). The specific Phase Theory predictions arising from coherence-scale physics (corrections to Coulomb's law, force law modifications at ultra-short distances) are therefore not directly testable in the near future. The testable predictions of Phase Theory (Sections 34–35) arise from lower-energy effects: gravitational corrections at the sub-millimeter scale, near-horizon effects in black hole physics, and precision coupling-constant measurements.

37.4 Conceptual Tensions

Phase Theory faces two unresolved conceptual tensions. (a) *The role of the observer*: the admissibility functional $A[\Phi]$ is defined as a functional of the global phase configuration Φ . But in practice, an observer can only access local information (Axiom 4, coherence limitation). How does the local observer's perception of forces relate to the global admissibility gradient? This tension is related to the measurement problem in quantum mechanics and is discussed in the measurement theory series (Papers 27–29). (b) *The origin of the initial phase configuration*: Phase Theory provides the dynamics of Φ (minimization of I) but does not specify the initial condition $\Phi_{t=0}$ (the state of the universe at its "beginning"). The initial conditions problem is deferred to the cosmology series (Papers 33–37).

38. Future Directions

38.1 Planned Sequel Papers

The results of the present paper open several lines of investigation, which are planned as sequel papers in the Phase Theory series. *Paper 43: Post-Newtonian Phase Corrections* — a systematic post-Newtonian expansion of Phase Theory gravitational force laws to 3PN order, with predictions for the phase-theoretic corrections to binary merger waveforms. *Paper 44: Gauge Theory from Phase Topology* — a derivation of the full Yang-Mills gauge theory structure (field strengths, covariant derivatives, gauge transformations) from the topological properties of the phase substrate, without taking the gauge group as a given. *Paper 45: Force Unification Scale* — a precise computation of the phase unification scale Λ_{PU} using Phase Theory renormalization group equations, including all corrections from the metric phase mode and from discrete DAG effects at the Planck scale.

38.2 Broader Research Program

Beyond these specific sequel papers, the Phase Theory force emergence program points toward several broader research directions. (a) *Quantum Phase Theory*: the present paper treats Phase Theory at the classical level (minimization of I without quantum fluctuations). A full quantum treatment — Phase Theory path integral with all fluctuations of Φ — will produce quantum corrections to force laws, including the Lamb shift (from one-loop corrections to the Coulomb potential) and the anomalous magnetic moment of the electron. This quantum Phase Theory is under development in Papers 50–55. (b) *Phase Cosmology*: the force laws derived here have implications for the early universe, where the phase substrate was in a high-inconsistency state and the force structure was dramatically different from today. In particular, the electroweak phase transition and the QCD phase transition are, in Phase Theory, transitions in the topology of the admissibility landscape — changes in the basin structure that reconfigure the dominant force laws. (c) *Information and Force*: Axiom 4 (coherence limitation) implies a deep connection between information, entropy, and force in Phase Theory. The force laws derived here can be interpreted as gradient flows of a phase-geometric entropy functional, connecting Phase Theory to the holographic and entropic approaches to gravity [39–41].

39. Conclusion

We have presented a comprehensive derivation of all effective force laws from the geometry of the admissibility landscape in Phase Theory. The central conclusion is that force, as a concept, is emergent and approximate — a useful macroscopic description of defect motion along admissibility gradients in a phase-geometric substrate, valid in the smooth-variation, weak-field, non-relativistic limit and corrected systematically as these conditions are relaxed.

The main results of this paper are as follows. **Force as admissibility gradient (Section 7):** the effective force on a topological defect is $F_{\text{eff}} = m\nabla A$, where A is the admissibility functional and m is the inertial mass. Newton's second law is derived, not postulated. **Inverse-square law from geometric dilution (Section 10, Proposition 42.5):** in three emergent spatial dimensions, the admissibility gradient from a localized isotropic defect scales as r^{-2} . This is the universal mechanism underlying both Coulomb's law and Newton's gravity. **Coulomb's law (Section 11):** derived from the $U(1)$ sector of the admissibility gradient, with the Coulomb constant $K_{U(1)} = (4\pi\epsilon_0)^{-1}$ identified as the $U(1)$ phase stiffness. **Newton's gravity (Section 12):** derived from the metric sector of the admissibility landscape, with $G = \alpha/(8\pi K_{\text{metric}})$. Gravity is universally attractive because phase-inconsistency energy is always positive. **Attraction and repulsion from topology (Section 14):** complementary topological invariants attract; identical invariants repel — in the perturbative regime. **Force range from phase mode mass (Section 15):** massless modes give long-range (Coulomb/Newton) forces; massive modes give Yukawa (exponentially screened) forces, with range $\lambda = 1/m_\phi$. **Newton's three laws reinterpreted (Section 17):** all three Newtonian laws are consequences of phase geometry — inertia from admissibility basin structure, $F = ma$ from variational dynamics, action-reaction from symmetry of the admissibility functional. **Lorentz force from gauge invariance (Section 20, Theorem 42.10):** $f = q(E + v \times B)$ is uniquely determined by $U(1)$ gauge invariance of the admissibility functional applied to moving charged defects. **No-Force-Carrier Theorem (Section 27, Theorem 42.1):** no phase substrate degree of freedom is exchanged between defects during a force interaction. Forces are purely geometric. Virtual particles are perturbative bookkeeping devices. **Post-Newtonian gravity (Section 23):** all four classical tests of GR (perihelion precession, light deflection, gravitational time dilation, gravitational waves) are recovered exactly. **Force unification (Section 29):**

all four forces unify at $\Lambda_{\text{PU}} \sim 10^{15-16}$ GeV from the same geometric mechanism.

Predictive deviations (Sections 34-35): five specific experimental predictions distinguished from Standard Model and GR are enumerated.

The philosophical conclusion that emerges from this technical analysis is clear and simple, and we state it as explicitly as possible: *forces are shadows cast by phase geometry*. When a charged particle "pulls" another toward it, nothing is being pulled; no photon is emitted and absorbed; no mediating entity passes between them. The phase substrate simply prefers configurations in which complementary winding structures are proximate, and this preference — a geometrical fact about the admissibility landscape — manifests in the language of emergent mechanics as a force. This is not a metaphor. It is the precise content of Theorem 42.1 (No Force Carrier Theorem) and Proposition 42.4 (Effective Force Law). Nothing pulls. Phase prefers.

Appendix A: Proof of Proposition 42.5 (Inverse-Square Law from Geometric Dilution)

We give here a more complete proof of Proposition 42.5, including the non-isotropic case and the relation between the effective admissibility charge Q_{eff} and the topological winding number of the defect.

Setup. Let D be a localized topological defect at the origin with winding number $n \in \pi_k(T)$ in some sector of T with phase stiffness K_{sector} . The phase configuration in the exterior region $r \gg L_{\text{defect}}$ satisfies the linearized Euler-Lagrange equation:

$$K_{\text{sector}} \nabla^2 \varphi_{\text{sector}}(r) = 0 \quad (r > L_{\text{defect}}), (\text{A.1})$$

with boundary condition: $\oint_{S_r} K_{\text{sector}} (\nabla \varphi_{\text{sector}}) \cdot dS = n K_{\text{sector}} \Omega_D$ (constant for all $r > L_{\text{defect}}$), where Ω_D is a dimensional constant determined by the normalization of the winding number and the geometry of T .

Flux conservation. By Green's theorem applied to equation (A.1) in the annular region $L_{\text{defect}} < r_1 < r < r_2$:

$$\oint_{S_{r_2}} K_{\text{sector}} (\nabla \varphi) \cdot dS = \oint_{S_{r_1}} K_{\text{sector}} (\nabla \varphi) \cdot dS = Q_{\text{eff}} = n K_{\text{sector}} \Omega_D. \quad (\text{A.2})$$

Equation (A.2) is the admissibility Gauss law: the total admissibility flux Q_{eff} is conserved over any sphere enclosing the defect.

Gradient magnitude. For an isotropic defect, $\nabla \varphi = |\nabla \varphi|(r) \hat{r}$ by symmetry. The surface integral in (A.2) over S_r gives:

$$K_{\text{sector}} |\nabla \varphi|(r) \times 4\pi r^2 = Q_{\text{eff}}, \quad (\text{A.3})$$

hence $|\nabla \varphi|(r) = Q_{\text{eff}}/(4\pi K_{\text{sector}} r^2)$, and the admissibility gradient:

$$|\nabla A_{\text{sector}}|(r) = K_{\text{sector}} |\nabla \varphi|(r) = Q_{\text{eff}}/(4\pi r^2). \quad (\text{A.4})$$

Non-isotropic case. For a non-isotropic defect with winding structure that has angular dependence $Y_{\ell m}(\theta, \varphi)$, the general solution to (A.1) in the exterior is a superposition of multipole terms:

$$\varphi_{\text{sector}}(r, \theta, \varphi) = \sum_{\ell, m} (A_{\ell m}/r^{\ell+1}) Y_{\ell m}(\theta, \varphi). \quad (\text{A.5})$$

The monopole term ($\ell = 0$) gives $A_{00}/r = Q_{\text{eff}}/(4\pi r)$ and dominates at large r , recovering the inverse-square force from its gradient. Higher multipoles ($\ell \geq 1$) decay faster than $1/r$ and give force corrections at order $r^{-(\ell+2)}$ — subdominant at large separations. This justifies the isotropic treatment of Proposition 42.5 for the leading-order force law. $\square$

Appendix B: Proof of Theorem 42.1 (No Force Carrier Theorem — Full Version)

We provide here a more rigorous version of the proof of Theorem 42.1, formulated in the language of elliptic PDE theory. Let M_{em} be the emergent spacetime manifold with metric $g_{\mu\nu}$. Let D_A and D_B be two topological defects at positions x_A and x_B , with respective source terms $\rho_A(x)$ and $\rho_B(x)$ (supported in neighborhoods of x_A and x_B of diameter L_{defect}). The interaction phase configuration $\delta\Phi_{AB}$ satisfies:

$$L[\delta\Phi_{AB}](x) = 0 \quad x \notin \text{supp}(\rho_A) \cup \text{supp}(\rho_B), (B.1)$$

where $L = D_\mu(K^{\mu\nu} D_\nu)$ is the linearized phase consistency operator (a second-order elliptic operator). The boundary conditions are: $\delta\Phi_{AB} \rightarrow 0$ as $|x| \rightarrow \infty$, and $[\delta\Phi_{AB}]$ satisfies matching conditions across $\partial(\text{supp } \rho_A)$ and $\partial(\text{supp } \rho_B)$ determined by the presence of the defects.

By the Lax-Milgram theorem (in the Euclidean/elliptic signature) or the retarded Green's function construction (in the Lorentzian signature), equation (B.1) with these boundary conditions has a unique solution $\delta\Phi_{AB}(x) = G_A(x) + G_B(x)$, where $G_A(x) = \int G(x,y) \rho_A(y) d^4y$ is the Green's function solution sourced by ρ_A alone, and similarly for G_B . This solution is determined entirely by the source configurations ρ_A and ρ_B and the Green's function of L — it does not require any "emission" or "propagation" from A to B in the sense of a local, causal energy transfer from the source defect to the field and then to the test defect. The Green's function $G(x,y)$ is a global geometric object (a property of the entire manifold M_{em}), not a particle being emitted by A and absorbed by B. This completes the formal proof that the force is geometric and involves no exchange of substrate degrees of freedom. $\square$

Appendix C: Complete Catalog of Effective Potentials in Phase Theory

Name	T Sector	Effective Potential $V(r)$	Force $F(r)$	Range	Origin in Phase Theory
Coulomb	U(1)	$q_1 q_2 / (4\pi\epsilon_0 r)$	$q_1 q_2 \hat{r} / (4\pi\epsilon_0 r^2)$	∞	Massless U(1) mode, geometric dilution
Newton	Metric	$-Gm_1 m_2 / r$	$-Gm_1 m_2 \hat{r} / r^2$	∞	Massless metric mode, geometric dilution, always attractive
Yukawa (weak)	SU(2)	$-g^2 e^{-m_W r} / (4\pi r)$	$g^2 (1 + m_W r) e^{-m_W r} \hat{r} / (4\pi r^2)$	$1/m_W$	Massive SU(2) mode, exponential decay
Linear confinement	SU(3)	σr	$-\sigma \hat{r}$	$\sim 1 \text{ fm}$ (then break)	Flux tube in SU(3) sector, 1D dilution
Cornell (QCD)	SU(3)	$-\alpha_s / r + \sigma r$	$\alpha_s \hat{r} / r^2 - \sigma \hat{r}$	Confined	Sum of Coulomb + linear in SU(3)
1PN gravity	Metric	$-Gm_1 m_2 / r - 3G^2 m(m_1 + m_2) / (2r^2 c^2)$	GR correction to Newton	∞	Nonlinear metric admissibility curvature
van der Waals	U(1) (2nd order)	$-C_6 / r^6$	$6C_6 \hat{r} / r^7$	$\sim \text{nm}$	2nd-order admissibility interaction ; induced

Name	T Sector	Effective Potential $V(r)$	Force $F(r)$	Range	Origin in Phase Theory
					dipoles
Casimir (plates)	All (vacuum)	$-(\pi^2/720)(\hbar c/d^3)/A$	$-(\pi^2/240)(\hbar c/d^4)/A$	$\sim \mu\text{m}$	Restriction of vacuum phase fluctuations
Harmonic (bound)	Any (near basin)	$(1/2)k(r-r_0)^2$	$-k(r-r_0)\hat{r}$	$\sim r_0$	Quadratic approximation near admissibility basin
Dark Coulomb	T_{dark}	$q_{\text{dark}}^2/(4\pi K_{\text{dark}}r)$	$\sim 1/r^2$	∞ (if massless)	Massless dark U(1) mode, geometric dilution

Table C.1. Complete catalog of effective potentials in Phase Theory.

Appendix D: Notation and Conventions

We use natural units $\hbar = c = 1$ throughout unless otherwise specified. Spacetime indices are Greek ($\mu, \nu, \alpha, \beta = 0, 1, 2, 3$); spatial indices are Latin ($i, j, k = 1, 2, 3$). The spacetime metric signature is $(-, +, +, +)$. The emergent metric is $g_{\mu\nu}(x)$, its determinant $g = \det(g_{\mu\nu})$, and the inverse metric $g^{\mu\nu}$. The volume element is $d\mu(x) = \sqrt{(-g)} d^4x$. The covariant derivative compatible with the emergent metric is ∇_μ ; the gauge-covariant derivative is $D_\mu = \nabla_\mu - ig A_\mu(x)$ (where A_μ is the relevant gauge connection component). Phase configurations are denoted Φ (bold) for the full target-manifold map or φ (italic) for a specific component. Topological winding numbers are n, q, k (context-dependent). The admissibility functional is $A[\Phi]$ (square brackets for functional arguments; round brackets for function arguments). The phase-

inconsistency functional is $I[\Phi]$. Defect positions in emergent space are $\mathbf{x}$, $\mathbf{x}_A$, $\mathbf{x}_B$ (bold or with subscripts). The coherence length is ξ ; the DAG lattice spacing is $\ell_P \sim 10^{-35}$ m. The phase unification scale is $\Lambda_{PU} \sim 10^{15-16}$ GeV.

Appendix E: Glossary of Phase Theory Terms

Term	Definition	First defined
Phase configuration Φ	Map from base space M to target manifold T ; the sole fundamental constituent of reality	Paper 1 [16]
Phase-inconsistency functional $I[\Phi]$	Functional on phase configurations measuring global phase inconsistency; minimized by realized Φ^*	Paper 5 [12]
Admissibility functional $A[\Phi]$	$A_{\max} - I[\Phi]$; maximized by realized Φ^* ; quantifies "phase preferability"	Def. 42.1 (this paper)
Admissibility gradient ∇A	Rate of change of A with respect to defect position; the Phase Theory analog of force	Def. 42.6 (this paper)
Admissibility landscape	The function A as a function of defect configuration space; encodes all force phenomenology	Section 8 (this paper)
Topological defect (particle)	Stable, localized region with non-trivial topological charge in $\pi_k(T)$; the Phase Theory model of a particle	Paper 9 [13], Def. 42.3
Admissible transport	Defect motion that preserves topological	Def. 42.5 (this paper)

Term	Definition	First defined
	sector and satisfies consistency equation in exterior region	
Inertial mass m	Second variation of I with respect to defect position; resistance to admissible phase transport	Paper 38 [14], Def. 42.7
Effective force F_{eff}	$m\nabla A$; emergent approximation to defect motion in smooth admissibility landscapes	Prop. 42.4 (this paper)
Geometric dilution	Mechanism by which admissibility gradient decreases as $1/r^2$ in 3D due to spreading over a sphere of area $4\pi r^2$	Section 10 (this paper)
Phase mode mass m_ϕ	Mass of small-amplitude fluctuation in T sector, determined by curvature of V_{top} at vacuum; determines force range	Def. 42.10 (this paper)
Phase unification scale Λ_{PU}	Energy scale at which all T -sector couplings unify; estimated 10^{15-16} GeV	Section 29 (this paper)
Reconfiguration event	Discrete change in defect topological invariant during interaction; not captured by force description	Paper 41 [15]
Coherence length ξ	Scale over which phase coherence is maintained; provides natural UV cutoff; $\lesssim 10^{-19}$ m	Paper 5 [12]
Update DAG	Directed acyclic graph encoding causal order of phase updates on	Paper 5 [12], Axiom 2

Term	Definition	First defined
	base space M ; gives rise to spacetime	

Table E.1. *Glossary of key Phase Theory terms used in this paper.*

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